The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

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Abstract

Objective. Temporal-interference (TI) stimulation applies two high-frequency currents whose amplitudes beat at a low difference frequency, a proposal for improving the spatial selectivity of non-invasive electrical stimulation at depth. The field carries no component at the beat, so a beat-frequency response requires a nonlinearity. We ask which features of the response follow from population nonlinearities and recurrent dynamics, and where the detector's own dynamics matter. **Approach.** We compare two modeled detectors, a memoryless firing-rate sigmoid (Jansen–Rit column) and an exact quadratic integrate-and-fire mean field with its spiking network, each feeding a recurrent network near a Hopf bifurcation, driven by genuine two-tone fields. **Main results.** Both detectors generate a difference-frequency component that scales with the product of the two carrier polarizations, and the network amplifies it as $1/\gamma$ in the damping, tunable by connectivity at fixed generated drive. The detectors differ: the sigmoid mixes through its curvature and reverses sign at its inflection; the dynamic model at a fast carrier mixes the carrier-frequency voltage excursion and reads it out through the population slope, the crossover lying at the population's own frequency. A steady-state rate curve therefore does not determine the response to a fast carrier. The beat-locked modulation of the rate grows toward the bifurcation far faster than the mean rate, in the mean field and in a network of 8000 neurons per population. TI and a direct in-band drive share the amplifier, so resonance alone gives TI no efficiency advantage. **Significance.** The population is a tuned receiver whose detector is model-dependent. Carrier access, nonlinear beat generation and circuit tuning should be measured separately, and in the local square-law description the generated beat depends on the product of the two carrier inputs rather than on the envelope excursion, which bears on dose maps.

Keywords: temporal interference; non-invasive neuromodulation; transcranial stimulation; neural mass model; amplitude demodulation; Hopf bifurcation; cross-frequency coupling; criticality.

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1 Introduction

Reaching a deep brain target without stimulating the tissue above or around it is a long-standing goal of non-invasive neuromodulation, and temporal-interference (TI) stimulation [1] is its most prominent recent proposal. TI applies two sinusoids at f_1 and $f_2 = f_1 + \Delta f$ through separate electrode pairs (typically $f_1 \sim$ 1–2 kHz, $\Delta f \sim$ 1–100 Hz). Where the two fields overlap, their superposition is an amplitude-modulated (AM) signal: a carrier near f_1 whose envelope beats at the difference frequency Δf. The appeal is spatial: the envelope amplitude peaks in a focal, steerable region at depth, while superficial tissue sees only the (putatively inert) kHz carrier—a promise of non-invasive, focal, deep stimulation that explains the intense interest.

A mechanistic question remains: how is a response at the beat frequency generated? Write the field as the sum of the two applied tones,

$$s(t) = \varepsilon_1 \cos(\omega_1 t) + \varepsilon_2 \cos(\omega_2 t), \qquad \omega_{1,2} = 2\pi f_{1,2}, \quad \Omega = \omega_2 - \omega_1 = 2\pi\,\Delta f, \tag{1}$$

where $\varepsilon_{1,2}$ are the amplitudes of the two tones at the target (refined to post-coupling polarizations in §2.2), $\Delta f = f_2 - f_1$ is the envelope (beat) frequency and $f_c = (f_1 + f_2)/2$ the carrier. Its spectrum has support *only* at ω_1 and ω_2 (Fig. 3): there is no energy at the envelope frequency Ω. A linear, time-invariant (LTI) system maps a sinusoid to a sinusoid of the same frequency; it can attenuate or phase-shift the two lines but can *never* synthesize a new line at Ω. The membrane's passive RC low-pass—often invoked as "the neuron follows the slow envelope"—therefore cannot by itself demodulate TI, a point now widely accepted [2]. Demodulation requires a nonlinearity. This is the founding lesson of AM radio [3]: a receiver cannot recover the message with linear components alone; it needs a nonlinear element (a square-law device or a diode) to shift spectral energy down to *baseband*—the low-frequency band near 0 Hz where the message lives—followed by a tuned filter to select the station and a low-pass to read the envelope.

Where, then, is the nonlinearity in the brain? The leading biophysical account places the nonlinearity in single-neuron ion channels: Mirzakhalili et al. [2] show in Hodgkin–Huxley and MRG-type mammalian myelinated axon models that TI works through nonlinear, ion-channel-mediated rectification, in which asymmetric voltage-gated conductances rectify the carrier and, after membrane low-pass, follow the envelope—the textbook rectify-then-filter detector. Other studies report that TI thresholds inherit the carrier's frequency dependence, suggesting a mechanism shared with direct kHz stimulation (conduction block, onset response) rather than a dedicated envelope channel [4, 5], and in some peripheral preparations the response is not driven by envelope extraction at all [5]. In vivo, TI reliably alters spike *timing* but not firing rate in the primate brain, and is much weaker than conventional low-frequency stimulation under human-relevant conditions [6]. Recent slice work finds that pyramidal and parvalbumin (PV) interneurons respond differently and that most pyramidal TI responses disappear when synaptic transmission is blocked, leading Caldas-Martinez et al. to conclude that TI is "largely a network phenomenon" [7]; computational network models point the same way [8, 9], and current reviews describe the mechanistic account as having moved from passive filtering to ion-channel rectification to network-mediated inhibition [10]. In peripheral nerves and isolated neurons at high intensity, the kHz carrier itself stimulates and thresholds follow the carrier rather than the envelope [4, 11].

We ask which features of the TI response follow from standard population nonlinearities and recurrent dynamics, and where additional biophysics is required. Two ingredients already in standard cortical models carry the argument: a firing-rate *sigmoid*, whose nonzero curvature σ'' makes it a square-law demodulator, and a *second-order-synapse network* whose synaptic kinetics filter the carrier away while a weakly-damped *stable focus* resonantly amplifies the recovered beat component at the network's natural frequency (Fig. 1). The amplification is a property of the focus nearing criticality—here the Hopf that opens the alpha rhythm—and is not tied to one parameter. (A saddle-node on an invariant circle gives the column a separate, slower rhythm, but through excitability with a diverging period rather than a sharp focus resonance, so its amplification is

of a different kind.) Where existing accounts locate the nonlinearity in cellular excitability, we compare two modeled detectors: the coarse-grained population sigmoid, whose curvature detects envelopes for carriers slow compared with the population's own dynamics, and the exact mean field of a quadratic integrate-and-fire population, in which the cell's voltage quadratic mixes a fast carrier and the population supplies the read-out. The two agree on quadratic scaling and disagree on the coefficient and its state dependence. What is common, and what this paper establishes, is nonlinear difference-frequency generation followed by network tuning, carrier-independent at fixed polarization and frequency-selective: a candidate network-level route to TI complementary to the disinhibitory network effect (PV-interneuron silencing) reported by Caldas-Martinez et al. [7]; both place TI's action in the population rather than the single cell. We develop it first in a single Jansen–Rit column, analyzable in closed form, then in an exact next-generation mean field and the spiking network beneath it, where the nonlinearity is derived rather than assumed; a two-band laminar model (LaNMM), whose own alpha and gamma loops bring it closer to cortex, is treated as an extension at low carriers.

This work builds on a line of research in which neural-mass models implement AM-radio principles *internally*, for the brain's own signals. In our cross-frequency-coupling account of predictive coding [12], the LaNMM realizes a columnar Comparator through sigmoid-enabled signal-to-envelope (SEC) and envelope-to-envelope (EEC) couplings, where error evaluation is literally demodulation—the sign-reversed prediction "un-modulates" the carrier. The Hierarchical Amplitude Modulation (HAM) framework [13] generalizes this to nested envelopes with constant-Q (log-spaced) bands, a candidate origin of $1/f$ spectra—again through the sigmoid's quadratic cross-term. The present work turns this machinery outward: if the cortex is an AM transceiver for its own rhythms, it can also be an AM receiver for an external carrier.

Throughout we distinguish the nonlinear generation of a difference-frequency drive from the network's response to it. The memoryless sigmoid provides the reference calculation; its steady-state curvature need not describe detection at high carrier frequencies, and the exact mean field shows where it fails. The aim is to identify which predictions are shared across the models, which depend on the detector and the input convention, and what additional biophysics connects the results to transcranial stimulation.

2 Theory

2.1 Neural mass models and the Jansen–Rit circuit

A neural mass model (NMM) collapses the activity of a large population ($\sim 10^4$) of similar neurons into a handful of population-averaged variables [14, 15]. Two elements recur in the Jansen–Rit-type models considered first. First, each synapse acts as a *linear filter* $\hat{K}$ that converts an incoming presynaptic firing rate $r(t)$ into a postsynaptic potential (PSP) $x(t) = \hat{K}[r]$; equivalently $r = \hat{L}[x]$ with $\hat{L} = \hat{K}^{-1}$ a differential operator whose few coefficients set the synapse's gain, decay time and delay [14]. In the Jansen–Rit (JR) family the kernel is the "alpha function" $h(t) = Aa\, t\, e^{-at}$ (excitatory; B, b for the inhibitory synapse), i.e. the second-order operator $\hat{L} = (Aa)^{-1}(\mathrm{d}/\mathrm{d}t + a)^2$, so each PSP obeys $\ddot{x} + 2a\dot{x} + a^2 x = Aa\, r(t)$. Second, the PSPs converging on a population sum to a mean membrane potential $v = \sum_s x_s$, which a static, saturating *transfer function*—the sigmoid—turns back into a population firing rate [16],

$$\sigma(v) = \frac{2e_0}{1 + e^{\,\rho(v_0 - v)}}, \tag{2}$$

with maximal firing rate $2e_0$, half-maximal at the inflection v_0, and slope set by ρ; it saturates because firing rates are bounded [14]. The sigmoid is the only nonlinearity in the model, and in this model it is what makes (de)modulation possible: it is memoryless, so whatever enters its argument, a kHz carrier included, is mapped to a rate instantaneously. That is an idealization. The curve is the ensemble average of single-neuron input–output relations for slowly varying input, so its

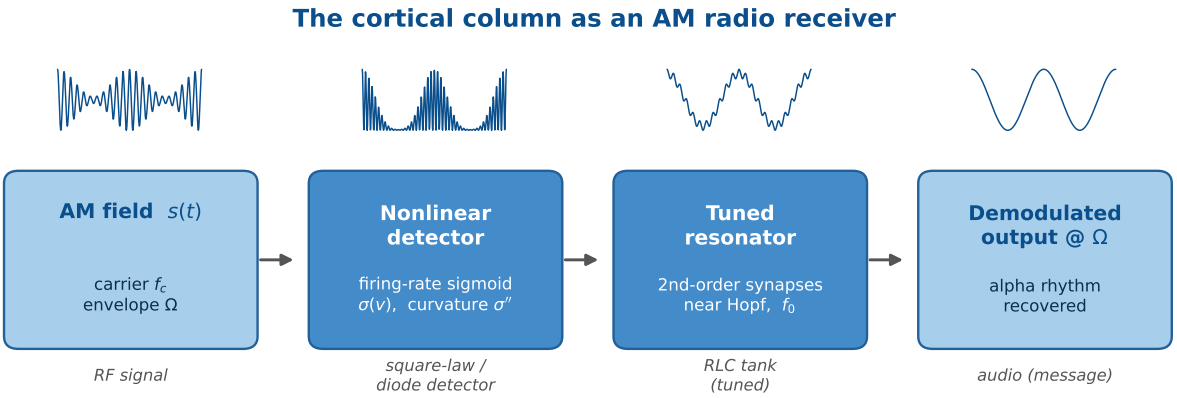


Figure 1: The cortical column as an AM radio receiver (schematic). An AM field (carrier f_c, envelope Ω) reaches a nonlinear detector (the firing-rate sigmoid, curvature σ''), whose output passes through a tuned resonator (the second-order-synapse network near a Hopf, natural frequency f_0), recovering a beat-frequency component as an oscillation at Ω. The analogy is functional rather than circuit-identical. In a classic AM receiver the tuned resistor–inductor–capacitor tank sits *before* the detector and selects the carrier [3]; here the sigmoid generates the baseband beat component, and a *baseband* resonator downstream of it selects and amplifies modulation frequencies near the column's own rhythm, so TI "broadcasts" to whichever columns are tuned to its beat frequency. A carrier-selective stage upstream of the detector does appear in the laminar model (§E).

curvature σ'' summarizes cellular nonlinearities (thresholds, spike generation, noise) smoothed by the spread of states across the population. Whether a fast carrier reaches that curve unchanged is a question the memoryless model cannot answer; the exact mean field below (Eq. (4)) can, and §2.3 shows that it does not. The network supplies the resonant *amplification* (§2.5). Frequency-selective resonance can also be intrinsic to single cells (e.g. I_h and I_M currents [17]); we locate it instead in the network, where second-order synapses on a loop near criticality generate it even when the cells do not [18, 19].

Assembled into a network, these two elements give the NMM1 transfer-function form of [14] (their Eq. 6.5): for each synapse $m \leftarrow n$,

$$\hat{L}_{m\leftarrow n}[u_{m\leftarrow n}] = C_{m\leftarrow n}\, r_n, \qquad v_m = \Lambda[\hat{E}] + \sum_{n:\, C_{m\leftarrow n}\neq 0} u_{m\leftarrow n}, \qquad r_m = \sigma(v_m), \tag{3}$$

with $u_{m\leftarrow n}$ the PSP produced at synapse $m \leftarrow n$, $C_{m\leftarrow n}$ the connectivity, and $\Lambda[\hat{E}]$ any *direct*, non-synaptic perturbation of the membrane. An applied field enters precisely here, as the additive term $\Lambda[\hat{E}]$ in the sigmoid's argument, alongside the synaptic PSPs—the starting point for §2.2. The concrete equations and parameters of the models simulated are collected in App. B, in this formalism.

This becomes exact one rung up the modeling ladder [14]. For a network of all-to-all coupled quadratic integrate-and-fire (QIF) neurons with Lorentzian-distributed excitabilities, the Ott–Antonsen ansatz yields a *closed, exact* mean field for the population firing rate r and mean potential v—the next-generation (Montbrió–Pazó–Roxin, MPR) mass [20], completed with the second-order synapses of neural-mass models in the exact "NMM2" theory [21, 22]:

$$\dot{r} = \frac{\Delta}{\pi} + 2rv, \qquad \dot{v} = v^2 + \bar{\eta} - \pi^2 r^2 + I(t), \tag{4}$$

with the applied current $I(t)$ entering linearly, the common background excitability $\bar{\eta}$ (the bifurcation parameter), and the heterogeneity half-width Δ setting the smoothing. Here the nonlinearity

is *derived, not postulated*: the quadratic v^2 term, inherited from the single-neuron QIF dynamics, mixes whatever the membrane potential carries, and the applied current enters the voltage equation rather than a rate map. For a constant input Q the population settles on the rate $r = F(Q)$,

$$F(Q) = \frac{1}{\pi}\sqrt{\frac{Q + \sqrt{Q^2 + \Delta^2}}{2}}, \tag{5}$$

a sigmoid-like curve with a convex foot, an inflection at $Q = \Delta/\sqrt{3}$ and an unbounded concave rise [22]; it is the slow-input limit of the dynamics, not a logistic. The Jansen–Rit sigmoid is the heuristic (NMM1) stand-in for such a curve. Both models mix two carriers quadratically, but their detector coefficients are not the same object: the memoryless sigmoid's is its curvature, the MPR population's depends on the carrier frequency through the voltage dynamics (§2.3). The same exact theory exhibits weak-field resonance and Arnold tongues that sharpen with self-coupling [22, 21], the near-Hopf amplification we exploit below.

What the transfer functions represent. In every model the demodulating object is a *population* input–output map rather than a passive somatic membrane element. In JR/LaNMM it is the sigmoid (2), the effective transfer of an excitable neuronal ensemble (mean PSP in, population firing rate out); in NMM2 the same role is played by the exact MPR quadratic (4). Its curvature is the mesoscopic trace of cellular nonlinearities (thresholds, spike generation, refractoriness, noise, saturation), so for inputs slow compared with the population's own dynamics, reading the demodulation off σ'' is legitimate at this scale; for a fast carrier the derived v^2 term mixes the voltage excursion and the population curve enters as the read-out gain (§2.3). The membrane low-pass does not contradict this: it bears on *access*—how the carrier reaches the nonlinearity—not on legitimacy. Because an applied field polarizes the membrane directly, in parallel, at every compartment (§2.2), access is set by the entry point, and the open biophysical question is which fast compartment (axon initial segment, node, or channel) still passes a kHz carrier to that nonlinear transfer (§2.6). Once the carrier is sampled there, the neural mass describes the population-level demodulation and near-Hopf amplification of the recovered beat component.

The single JR column couples a pyramidal population to one excitatory and one inhibitory interneuron population through this synapse+sigmoid motif, with the field entering the pyramidal sigmoid as $s(t) = \Lambda[\hat{E}]$ (§2.2) and the local field potential (LFP) proxy $v = y_1 - y_2$. Its equations (Eqs. (1)) and parameters (Table 2) are collected in App. B.

For the biophysical realization (§E) we use a second model built from the same ingredients: the laminar neural mass model (LaNMM) [12, 23], which embeds this synapse+sigmoid motif in a two-band cortical column—a deep Jansen–Rit-like subcircuit (pyramidal P_1 with spiny-stellate (SS) and somatostatin (SST) interneurons; intrinsic alpha, ∼10 Hz) coupled to a superficial PING-like (pyramidal–interneuron network gamma) subcircuit (pyramidal P_2 with PV interneurons; intrinsic gamma, ∼40 Hz). The single JR column lets us isolate and analyze the mechanism; the LaNMM then exhibits it in a column with its own fast/slow loops and two native resonances. Everything below applies to both: the field enters the sigmoid argument as a membrane polarization, and the recovered beat component drives a resonant circuit.

2.2 From applied field to a squared-envelope beat component

We model the applied TI field $E(t)$ (V/m) as a quasi-static perturbation of the membrane potential that the spike-generating nonlinearity reads. "Quasi-static" means we neglect electromagnetic propagation and induction (wavelengths $\gg$ head, frequencies far below tissue relaxation), so the field acts as an instantaneous polarization added inside the firing-rate map,

$$r(t) = \sigma\big(v(t) + s(t)\big), \qquad E(t) = E_1 \cos\omega_1 t + E_2 \cos\omega_2 t. \tag{6}$$

Unlike synaptic drive, an applied field does not enter through the synaptic kernel: it polarizes the membrane directly and simultaneously at every compartment—soma, dendrite, axon initial

segment (AIS), terminal—each with its own coupling and its own access filter. The perturbation $s(t)$ reaching the nonlinearity is therefore a sum over these parallel entry points, or "doors",

$$s(t) = \sum_c \boldsymbol{\lambda}_f^c \cdot \mathbf{E}(t), \qquad \lambda_f^c = \lambda_0^c \, |H_c(\omega_c)|, \tag{7}$$

where λ_0^c is the DC (quasi-static) coupling of door c—the weak-field polarization length $\delta V = \boldsymbol{\lambda}\cdot\mathbf{E}$ of transcranial-stimulation modeling [14, 24]—and λ_f^c is its effective coupling at the carrier, the DC value rolled off by that door's own filter H_c. The field is thus not only a somatic membrane perturbation, and which door dominates is set by the carrier: at DC the soma/dendrite has the largest coupling (the whole-cell dipole), but its filter cuts it at kHz while a fast door stays open, so λ_f^{soma} collapses and the dominant term crosses over (§2.6). The amplitudes reaching the nonlinearity are the post-coupling polarizations $\varepsilon_j = \lambda_f E_j$ of the open door (door index suppressed when only the open door matters), which is what Eq. (1) contains rather than the raw applied field. The main text takes the favorable direct limit in which a fast door (axonal or nodal) samples the carrier before appreciable attenuation, $\lambda_f \approx \lambda_0$, isolating the sigmoid+resonance principle; §2.6 restores the roll-off and shows the carrier dependence entering quadratically, $A_\Omega \propto |H_c(f_c)|^2$.

Expanding σ about an operating point v^* (a Volterra/Taylor expansion, as in [13]) gives $\sigma(v^* + s) \approx \sigma(v^*) + \sigma'(v^*)\, s + \frac{1}{2}\sigma''(v^*)\, s^2 + \dots$ The truncation holds only while $|s|$ is small against the sigmoid's voltage scale $1/\rho \approx 1.8$ mV. Human TI polarizations are $\sim 10^{-2}$ mV, over two orders below that, so the quadratic term is the leading nonlinearity throughout the regime this paper treats. Rodent TI is not in that regime: the fields are $\sim 10^3$ times larger, the sigmoid saturates, and the expansion fails. §5 returns to what that costs when rodent results are scaled to humans. Intuitively, the difficulty is that the field carries power only near the carrier ω_c, far from the slow envelope frequency Ω, so some operation must move power down to Ω. The *linear* term cannot: it passes the input lines straight through, reproducing the carrier and its two flanking *sidebands* (the lines at $\omega_c \pm \Omega$ created by the modulation), but still nothing at Ω. The *quadratic* term can: squaring the field multiplies the fast carrier by itself, and because $\cos^2$ has a constant (zero-frequency) part, the squaring folds power down to *baseband*—the low-frequency band near 0 Hz, where the slow envelope lives. This is *rectification*, what a diode does in an AM radio. The identity $2\cos\omega_1 t \cos\omega_2 t = \cos\Omega t + \cos(\omega_1 + \omega_2)t$ shows where the beat comes from: the cross term of s^2. Averaging over the carrier (the fast $\cos^2 \omega_j t$ terms average to $\frac{1}{2}$, the sum-frequency term to zero),

$$\big\langle \sigma(v^* + s) \big\rangle_c = \sigma(v^*) + \tfrac{1}{4}\sigma''(v^*)\,(\varepsilon_1^2 + \varepsilon_2^2) + \tfrac{1}{2}\sigma''(v^*)\,\varepsilon_1\varepsilon_2 \cos\Omega t + O(\varepsilon^4), \tag{8}$$

so a line appears at exactly the envelope frequency Ω, with amplitude

$$A_\Omega \simeq \tfrac{1}{2}\,\sigma''(v^*)\,\varepsilon_1\varepsilon_2, \tag{9}$$

where $\simeq$ means to leading order in the polarizations, here and in what follows. The same expansion leaves a constant (DC) term, $\frac{1}{4}\sigma''(\varepsilon_1^2 + \varepsilon_2^2)$, which any read-out of the mean rate must include.

What the quadratic term recovers is not the envelope but its square. Writing $A(t)$ for the instantaneous envelope of the two-tone field, $A^2(t) = \varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2\cos\Omega t$, the low-pass part of the squared signal is exactly

$$\mathrm{LPF}\big[s^2(t)\big] = \tfrac{1}{2}A^2(t) = \frac{\varepsilon_1^2 + \varepsilon_2^2}{2} + \varepsilon_1\varepsilon_2\cos\Omega t. \tag{10}$$

This is square-law detection in the strict sense: the population transduces A^2, not A. The distinction matters because A^2 is exactly sinusoidal at Ω whereas A is not—at a matched target ($\varepsilon_1 = \varepsilon_2 = \varepsilon$), $A(t) = 2\varepsilon|\cos(\Omega t/2)|$, a rectified cosine.

The amplitude-modulated surrogate. The dense perturbative sweeps below use the three-line waveform

$$s(t) = \varepsilon\,[\,1 + m\cos\Omega t\,]\cos\omega_c t, \qquad \omega_c = 2\pi f_c, \tag{11}$$

with modulation depth m, because it parameterizes a sweep by one amplitude. It is not the physical signal: it carries a line at ω_c that the two-tone sum lacks. The two agree at Ω to leading order (the surrogate gives $A_\Omega = \frac{1}{2}\sigma''\varepsilon^2 m$) and are *not* equivalent beyond it. For matched genuine tones $\mathrm{LPF}[s^2] = \varepsilon^2(1+\cos\Omega t)$, whereas for the $m=1$ AM surrogate $\mathrm{LPF}[s^2] = \varepsilon^2(\frac{3}{4}+\cos\Omega t + \frac{1}{4}\cos 2\Omega t)$: the same Ω amplitude, but a different DC term and a spurious second harmonic at 25% of the fundamental. Since that second harmonic is precisely the observable that distinguishes transduction classes (§2.4), every claim that turns on the waveform is run with genuine two tones: the demodulation demonstration (Fig. 3), the curvature and square-law controls (Fig. 6), the laminar maps (Figs. S1, S7) and the kHz demonstration. The AM surrogate is retained for the dense perturbative sweeps in (Ω, p) and (Ω, C), which read only the leading Ω component, and Fig. 6d verifies that the two agree there.

In the language of [13], the quadratic cross-term $s_1 s_2$ between a slow and a fast input (the second-order Volterra kernel) implements amplitude modulation/demodulation and the associated intermodulation lines. Demodulation is thus itself an intermodulation product: the line at Ω is the *difference frequency* of the two applied tones (ω_1 and ω_2), generated by the same quadratic term that—acting a second time, now on an intrinsic rhythm ω_0—produces the $\omega_0 \pm \Omega$ sidebands seen at the spiking level (Fig. S12b). One nonlinear mixer, applied in cascade. Equation (9) carries three predictions, all verified below: the recovered amplitude is quadratic in the drive ($A_\Omega \propto \varepsilon^2$, square-law detection); it follows the sigmoid curvature ($A_\Omega \propto \sigma''(v^*)$, hence zero at the inflection $v^* = v_0$); and, *at fixed post-coupling polarization* ε, it is independent of the carrier ω_c. This last prediction needs care: as a function of the *applied* field the response inherits the coupling $\varepsilon = \lambda_f E_0$, so the field required for a given effect rises with f_c (§2.6)—consistent with the experimental consensus. Carrier independence is thus a statement about the recovered beat component at fixed polarization, not about the applied field.

2.3 Static transfer versus dynamic mixing: the exact mean field

The sigmoid of Eq. (9) is memoryless, so the carrier traverses the curve and the curvature is the detector at any carrier frequency. A population of neurons is not memoryless: its rate relaxes and rings on its own timescales, and an applied current polarizes the membrane before any spike-generating nonlinearity sees it. The exact mean field makes the consequence explicit. Take a single MPR population (Eq. (4), $\tau = 1$, no coupling) with constant input Q and a weak current carrier $a\cos\omega t$ added to the voltage equation. Expanding to second order in a and averaging over the carrier, the rate acquires a constant shift $a^2\,C(Q,\omega)$ with

$$C(Q,\omega) = \frac{r^*\,[\omega^2 + 3d^2 - \nu^2]}{(d^2+\nu^2)\{[d^2+\nu^2-\omega^2]^2 + 4d^2\omega^2\}}, \qquad r^* = F(Q), \quad d = \frac{\Delta}{\pi r^*}, \quad \nu = 2\pi r^*, \tag{12}$$

where d is the population's relaxation rate and ν its ringing frequency (the linearization about the fixed point has eigenvalues $-d \pm i\nu$). Its two limits are the two detectors,

$$C(Q,\omega) \xrightarrow[\omega\to 0]{} \tfrac{1}{4}F''(Q), \qquad C(Q,\omega) \xrightarrow[\omega\gg\nu]{} \frac{F'(Q)}{2\omega^2}. \tag{13}$$

For a slow carrier the population tracks its steady-state curve and the static curvature rectifies, as in the Jansen–Rit calculation. For a fast carrier the voltage responds as an integrator, with an excursion $\sim a/\omega$ and a quarter-cycle lag; the v^2 term mixes that excursion into a constant, and the population converts the constant to a rate through its *slope*. The coefficient is then positive whatever the operating point and largest at the inflection, where F'' vanishes. Above the inflection the two limits have opposite signs and the coefficient changes sign at $\omega_z = \sqrt{\nu^2 - 3d^2}$ (Fig. 7a). The crossover sits at the population's own frequency, tens of hertz for the gamma-generating PING of §4.5; the 300 Hz carrier used there, and TI's kHz carriers, are on the fast side.

The same reasoning fixes what a two-tone field generates in the full eight-state PING of Eq. (24). In the fast-carrier limit each tone's voltage phasor is $a_j/(i\omega_j\tau_E)$, and the v_E^2 term produces a

difference-frequency source of magnitude $a_1 a_2/(\tau_E^2 \omega_1 \omega_2)$ in the τ_E-divided voltage equation, independent of the operating point; the response is that source passed through the network's susceptibility at Ω. Against the exact second-order response (full quadratic source and full resolvent) at four stable operating points, the fast-carrier source is within 1.1% at 300 Hz, 0.1% at 1 kHz and 0.03% at 2 kHz (Fig. 7b). Two consequences carry through the paper. Agreement in the ε^2 scaling between the sigmoid and the exact mean field does not mean the same detector: the coefficients and their operating-point dependence differ, and the sign reversal of Eq. (9) at the inflection is a property of the memoryless model. And carrier independence at fixed polarization survives in the dynamic model, with polarization now the carrier-frequency voltage excursion $a_j/(\omega_j \tau_E)$: at fixed current the generated source rolls off as $1/(\omega_1 \omega_2)$ from the membrane dynamics themselves.

2.4 Transduction class and its spatial and spectral signatures

A neural response proportional to the squared envelope is not equivalent to one proportional to the envelope. A detector implements some transduction $F(A)$ of the local envelope $A(t)$, and the two laws predict different amplitude scaling, harmonic content and spatial weighting, all three measurable. This section says how to tell them apart.

To make "class" concrete, compute what a single site emits at baseband. For a narrowband field $s(t) = A(t)\cos\omega_c t$ passing through a memoryless nonlinearity g, the low-frequency (baseband) output Y_{LF} is the average of g over one carrier cycle, $Y_{\mathrm{LF}}(A) = (2\pi)^{-1}\int_0^{2\pi} g(v^* + A\cos\theta)\,\mathrm{d}\theta$. Odd powers of A average away, leaving only even ones:

$$Y_{\mathrm{LF}}(A) = g(v^*) + \tfrac{1}{4} g''(v^*) A^2 + \tfrac{1}{64} g^{(4)}(v^*) A^4 + \cdots \tag{14}$$

A weak, smooth nonlinearity therefore transduces A^2 generically, whatever its detailed shape—which is what the sigmoid mechanism of §2.2 does, and why Eq. (9) is quadratic. Recovering A itself requires a qualitatively different detector: strong rectification. A full-wave rectifier gives $\mathrm{LPF}[\,|A\cos\omega_c t|\,] = (2/\pi)A$ and a half-wave rectifier $(1/\pi)A$, both linear in the envelope, as does a diode-and-capacitor peak detector. A hard threshold is not one of them: a step detector with offset δ transduces $\pi^{-1}\arccos(\delta/A)$ for $A > \delta$ and nothing below it, which is neither linear nor quadratic in A (App. D). Asymmetric channel rectification, threshold crossing and release events are candidate biological realizations of rectifiers of one kind or another. Under a linear downstream response and fixed coupling, the amplitude sweep below distinguishes a quadratic effective drive from one proportional to the envelope; the derivation of §2.2 assumes a smooth transfer, and this section asks what would show that assumption to fail.

The two classes are distinguishable at a single site. At a matched target, $A(t) = 2\varepsilon|\cos(\Omega t/2)|$, whose Fourier series $|\cos u| = \frac{2}{\pi} + \frac{4}{\pi}\sum_{n\geq 1} \frac{(-1)^{n+1}}{4n^2-1}\cos 2nu$ gives a harmonic ladder in ratios $A_{2\Omega}/A_\Omega = 1/5$, $A_{3\Omega}/A_\Omega = 3/35$ and $A_{4\Omega}/A_\Omega = 1/21$. The square-law output $\frac{1}{2}A^2 = \varepsilon^2(1+\cos\Omega t)$ contains only DC and Ω. For a mixed detector $F(A) = c_1 A + c_2 A^2$ the pre-network ratio is

$$r_{\mathrm{det}} \equiv \left|\frac{Y_{2\Omega}}{Y_\Omega}\right| = \frac{1/5}{1 + \frac{3\pi}{4}\, c_2\varepsilon/c_1}, \tag{15}$$

so measuring r_{det} estimates $c_2\varepsilon/c_1$ directly.

Two effects make the harmonic ratio usable only with care, and the second decides how to run the test. First, a smooth detector does generate a second harmonic beyond quadratic order, $Y_{2\Omega}/Y_\Omega \sim (c_4/c_2)\varepsilon^2$, so the square-law signature is a *vanishing* ratio in the weak-field limit, not an exact zero. Second, and more seriously, the recurrent network multiplies each harmonic by its own transfer gain, $r_{\mathrm{meas}} = r_{\mathrm{det}}\,|G(2\Omega)/G(\Omega)|$. A resonant column suppresses the second harmonic hard: driven at $\Omega = \omega_0$ with $\mathcal{Q} = 10$, a true 20% ratio reads as 0.7%, enough to make an envelope follower look square-law. The resonance is thus precisely what would hide the discriminator, so the test must be run *off* resonance: at $\Delta f \simeq f_0/4$ both lines sit on the flat

part of G, where $|G(2\Omega)/G(\Omega)| \approx 1.22$–$1.25$ across $\mathcal{Q} = 3$–30, or with G measured independently by a weak sweep of direct low-frequency stimulation (transcranial alternating-current stimulation, tACS) and deconvolved.

An amplitude sweep tests the scaling rather than one ratio. An envelope-following detector gives $Y_\Omega \propto \varepsilon$ with the harmonic ratio pinned at 1/5; a weak square-law detector gives $Y_\Omega \propto \varepsilon^2$ with a ratio growing as ε^2. The log–log slope of the envelope-locked response against field amplitude therefore separates the classes without requiring absolute calibration; the ε^2 control of §4.4 is that measurement, performed in the model.

Beyond harmonics and amplitude scaling, the transduction class fixes the spatial map. Define the excursion as $\mathcal{M}[y] \equiv \max_t y - \min_t y$ of a periodic quantity $y(t)$. The conventional TI envelope map is $\mathcal{M}[A] = 2\min(\varepsilon_1, \varepsilon_2)$, whereas the square-law detector responds to $\mathcal{M}[\frac{1}{2}A^2] = 2\varepsilon_1\varepsilon_2$, so the two are related pointwise by

$$\mathcal{M}\left[\tfrac{1}{2}A^2\right] = \max(\varepsilon_1, \varepsilon_2)\, \mathcal{M}[A]. \tag{16}$$

The square-law map is the conventional envelope-excursion map multiplied, at each point, by the dominant local polarization. It therefore does not preserve conventional TI focality. In typical montages, where the dominant carrier is substantially larger superficially than at depth, this reweighting degrades deep-to-superficial contrast: at $(\varepsilon_1, \varepsilon_2) = (10, 1)$ superficially against $(0.5, 0.5)$ at a matched deep target, $\mathcal{M}[A]$ favors the superficial point by $2\times$ and $\mathcal{M}[\frac{1}{2}A^2]$ by $40\times$. The magnitude of the degradation is not fixed by this algebra and must be evaluated from the actual vector field distribution; a sufficiently weak second carrier can suppress the superficial product instead. The product law is not an account of TI focality.

The selectivity this mechanism does supply is dynamical rather than spatial. Writing $R_\Omega(\mathbf{x})$ for the demodulated response at the envelope frequency Ω produced at location $\mathbf{x}$, and $G_\mathbf{x}(\Omega)$ for the local network susceptibility there,

$$R_\Omega(\mathbf{x}) \;\propto\; \sigma''(v^*_\mathbf{x})\, G_\mathbf{x}(\Omega)\, \varepsilon_1(\mathbf{x})\, \varepsilon_2(\mathbf{x}), \tag{17}$$

a product of three factors. Field overlap sets where a difference-frequency drive is generated at all; the operating point and the network susceptibility set which of the driven populations amplifies it. Resonance does not override an unfavorable field product, but it can compensate for one. This is the radio analogy taken literally: received signal strength times receiver tuning.

Equation (17) is written for scalar projected fields. In general the coupling is vectorial and compartment-specific, $\varepsilon_i(\mathbf{x}) = \boldsymbol{\lambda}\cdot\mathbf{E}_i(\mathbf{x})$, so the mixing product $(\boldsymbol{\lambda}\cdot\mathbf{E}_1)(\boldsymbol{\lambda}\cdot\mathbf{E}_2)$ is signed: its magnitude follows the product of the projected amplitudes, while its phase inverts with either the sign of σ'' or the relative sign of the two projections. The derivations below take the collinear, scalar case.

2.5 Amplification: the synaptic low-pass and the network resonance

The recovered beat component at Ω survives the loop only if the carrier is filtered out of it, and is then amplified by the network. For the carrier to leave, it need only be high enough that the synaptic low-pass suppresses it relative to the envelope, and "inert"—it must not itself fall in, or beat into, the network's responsive band, nor engage carrier-specific mechanisms (the ion-channel/kHz pathway, §2.6). Within those bounds the response is flat in f_c (Fig. 5, left). The synaptic suppression is concrete: each JR synapse applies the operator $\hat{L}_s$ with impulse response $h(t) = Aa\, t\, e^{-at}$, i.e. transfer function $H_s(\omega) = Aa/(i\omega + a)^2$, a critically-damped second-order low-pass with corner $a/2\pi \approx 16$ Hz. It passes the $\sim$10 Hz envelope ($|H_s(\omega)|/|H_s(0)| \sim 0.7$) while suppressing a 100 Hz carrier to $\sim$2.5% (Fig. 5, right), so the carrier is filtered out of the loop *after* the sigmoid has used it to demodulate. Thus two distinct mechanisms shape the closed-loop dynamics: the sigmoid generates a squared-envelope component at the beat frequency, whose

amplitude is inherently independent of the carrier, and the synaptic low-pass then removes the carrier itself from the spectrum.

What turns this recovered beat component into a large, frequency-selective response is the network. Linearizing the mass about its fixed point gives a leading eigenvalue pair $\lambda_\pm = -\gamma \pm i\omega_0$; near the Hopf the response is dominated by this critical mode, so the whole network behaves as a single weakly damped harmonic oscillator with damping γ and natural frequency ω_0 (the gain below is its one-mode, center-manifold reduction). Just below a supercritical Hopf, $\gamma \to 0^+$ and the mass is a sharply tuned (high quality-factor $\mathcal{Q} = \omega_0/2\gamma$) resonator; driving it with the demodulated beat component (9) at Ω gives the textbook resonance gain

$$G(\Omega) \propto \left[(\omega_0^2 - \Omega^2)^2 + (2\gamma\Omega)^2\right]^{-1/2}, \qquad G(\omega_0) \propto \frac{1}{2\gamma\omega_0}, \tag{18}$$

so the steady response $\sim A_\Omega\, G(\Omega)$ peaks at $\Omega = \omega_0$ (a Lorentzian in power) with a height that diverges as the Hopf is approached, within the weak-drive regime where the linear response holds. Here ω_0 is the imaginary part of the eigenvalue; the undamped frequency of the canonical oscillator differs from it by a term of order γ^2, immaterial near the Hopf. Just above the Hopf the mass is an autonomous limit cycle at ω_0, and the demodulated drive *entrains* it (an Arnold tongue), giving a response peak near ω_0 that grows with cycle amplitude.

The resonance amplifies but cannot detect. The universal Hopf normal form, whose nonlinearity is odd, responds to an in-band drive yet produces no line at Ω at any order in the field (App. C), so detection and amplification are separate requirements rather than two aspects of proximity to criticality.

2.6 Access and carrier independence: fast elements and the kHz carrier

Carrier independence (at fixed polarization) holds for the direct coupling (6); the access filter H_c of Eq. (7) restores the carrier dependence, and *which* entry point is open sets its corner. Take first the *somatic door*: the membrane (and dendritic cable) low-passes the field before the spike nonlinearity. (The demodulating σ'' is the *population* transfer; the door filter is the upstream *access* stage—this section asks only whether the carrier survives to it.) Modeling the somatic door as a first-order filter $H_m(\omega) = 1/(1 + i\omega\tau_m)$, the carrier amplitude reaching the detector is $\varepsilon_{\text{eff}}(f_c) = \varepsilon/\sqrt{1 + (\omega_c\tau_m)^2}$ (that is, $\lambda_f^{\text{soma}} E_0$ in the notation of Eq. (7)), and since $\Omega \ll \omega_c$ the three AM lines are attenuated almost equally, so

$$A_\Omega(f_c) \;\simeq\; \tfrac{1}{2}\,\sigma''(v^*)\, m\, \varepsilon_{\text{eff}}^2 \;=\; \frac{A_\Omega(0)}{1 + (\omega_c\tau_m)^2} \;\xrightarrow[\omega_c\tau_m \gg 1]{}\; A_\Omega(0)\,\frac{1}{(\omega_c\tau_m)^2}. \tag{19}$$

The demodulated response rolls off as $1/f_c^2$ above the membrane corner (Fig. S5). For a pyramidal soma/dendrite the passive time constant is $\tau_m \sim 10$–30 ms (human L2/3 pyramidal cells: $\sim$12 ms [25] and 16.5 ± 3.7 ms [26], which bracket the $\sim$16 ms we adopt for the soma, corner $\sim$10 Hz), so the carrier-power factor $|H_m|^2$ is only $\sim 10^{-4}$ at 1 kHz, $\sim 2.5 \times 10^{-5}$ at 2 kHz and $\sim 10^{-6}$ at 10 kHz. Passive somatic demodulation is therefore negligible at TI's kHz carriers—a four- to six-order-of-magnitude suppression. Equation (19) is the response computed through the somatic door: correct, but the door a kHz carrier finds shut.

In this access-filter model, at kHz the somatic door is shut and a faster one is open. A fast element (axon initial segment (AIS), node of Ranvier; $\tau \approx 0.2$ ms, effective corner $\sim$800 Hz) low-passes the same field only mildly ($|H|^2 \approx 0.39$ at 1 kHz, 0.14 at 2 kHz, still $\sim 6 \times 10^{-3}$ at 10 kHz; Fig. S5). This is not the parameter swap $\tau_m \to \tau_{\text{fast}}$ within one door but a distinct entry point of Eq. (7), in parallel with the soma rather than in series with it. A square-law nonlinearity there produces a real envelope-frequency current; because that component is generated at Ω *after* the access filter, no upstream pole can remove it, and it is exactly what the slow network reads. The per-neuron modulation is small, but two factors make it count: the envelope sits in the band the membrane

passes, and the population near a Hopf amplifies it by $G \sim 1/\gamma$ (Eq. (18)). Demodulation at fast elements, resonantly amplified at the population, is thus a candidate route for TI at realistic carriers, consistent with ion-channel rectification at the node [2] but adding network resonance; our direct-coupling demonstration is the $\tau \to 0$ idealization of this limit, and a direct kHz-carrier run (Fig. S9) confirms it.

Entry point (door)	Input	Access corner	Dominates when
synapse → dendrite → soma	rate r	∼16 Hz	normal operation
field → soma / dendrite	field ε	∼10 Hz	DC, slow tES
field → AIS	field ε	**∼800 Hz**	**kHz carriers (TI)**
Na_v activation gate (within σ)	—	∼1 kHz	sets the ceiling

Table 1: Entry points into the sigmoid argument of Eq. (7). The same field enters through several doors; corners are order-of-magnitude, and which door dominates is set by the carrier. The AIS door (bold) is the leading candidate for TI; the compartment is not yet settled. Presynaptic terminals polarize strongly but, as isopotential compartments, are no faster than the soma (§5).

The somatic and fast doors can be told apart only at kHz. At low frequency the AIS and soma are electrotonically one compartment, so "somatic" and "spike-initiating" potential are the same number and the two doors are indistinguishable: the neural-mass state variable is degenerate at DC. The degeneracy lifts as the AC space constant $\ell(\omega)$ contracts from its DC value ℓ_0,

$$\ell(\omega) = \ell_0 \, (1 + \omega^2 \tau_m^2)^{-1/4}, \tag{20}$$

by roughly tenfold at 1 kHz for $\tau_m = 16$ ms—to tens of μm, comparable to the AIS length itself. A kHz carrier is thus the regime in which the demodulating compartment becomes determinate; note that ε_{eff} as defined above absorbs only the membrane pole, not this contraction.

One lumped assumption then remains in the Jansen–Rit description: that σ is static; the exact mean field removes it (§2.3). A static σ is the $\tau_{\text{gate}} \to 0$ idealization of a spike-initiation nonlinearity whose Na_v activation gate has $\tau \approx 0.1$–0.2 ms, a corner near 1 kHz; above it σ'' itself rolls off by a factor $|H_{\text{gate}}(\omega_c)|^2$. This gate, not the membrane, sets the frequency ceiling of the mechanism, and it is measurable: a carrier sweep at fixed envelope frequency and field should roll off only in the kHz range, where the access corner (∼800 Hz) and the gate corner (∼1 kHz) combine, not at the ∼10 Hz corner of the somatic door.

Because transcranial electrical stimulation (tES)—and TI, a tES with a kHz carrier—is weak and *modulatory*, the fast-element route is a small, sustained bias of the most excitable compartments, not "spiking the AIS at the beat frequency." Since the applied field adds to endogenous drive, a weak bias near threshold shifts spike probability and timing through stochastic resonance [27, 28]. The population sigmoid (2) is itself a mean-field stochastic-resonance device—its smooth slope is the spread of thresholds and noise across the population—so the bias shifts the mean rate through σ' while the demodulated beat component emerges through the curvature σ''.

Both halves are state-dependent, detection through the operating point and amplification through $1/\gamma$, so TI efficacy should track brain state, not the stimulus alone. The 100 Hz carrier used throughout is not merely convenient: the membrane attenuation is mild even for moderately fast elements ($|H_m|^2 \approx 0.98$ at $\tau = 0.2$ ms, ∼0.1 at $\tau = 5$ ms), so a protocol with $f_c \sim 100$ Hz and Δf swept through a region's band is a clean near-term test. The trade-off is spatial: a 100 Hz carrier is less inert off-target than a kHz one (whose subthreshold carrier is the source of TI's focality), so 100 Hz probes the mechanism while kHz optimizes selectivity.

2.7 Operating-point dependence

Because $A_\Omega \propto \sigma''(v^*)$, demodulation is governed by the curvature at the operating point, and in particular by how far v^* sits from the sigmoid inflection (Fig. 6). The Population Excitation–Inhibition Index (PEIX) of [12] is a related but distinct quantity: the sigmoid gain $\sigma'(v^*)$ normalized to its maximum $\sigma'(v_0) = e_0\rho/2$ at the inflection,

$$\mathrm{PEIX}(v^*) \equiv \frac{\sigma'(v^*)}{\sigma'(v_0)} = \mathrm{sech}^2\Big[\tfrac{\rho}{2}\,(v^* - v_0)\Big] \in (0, 1], \tag{21}$$

a dimensionless read-out of where v^* sits, the side given by $\mathrm{sign}(v^* - v_0)$. PEIX thus tracks σ' (maximal at the inflection v_0), whereas the demodulation gain tracks σ'' (zero at v_0, peaked on the flanks, and sign-reversing across it): the two are complementary, and PEIX peaks exactly where the demodulation gain vanishes. This complementarity is a property of the memoryless detector. In the exact mean field at a fast carrier the rectified DC term follows the slope F' and is largest at the inflection (§2.3), so the relation between PEIX and TI efficacy is model-dependent; PEIX remains a measurable read-out of the operating point. In the Jansen–Rit detector, anything shifting v^* (neuromodulation, attention, arousal, or pathology such as psychedelics or PV-interneuron dysfunction in Alzheimer's disease) scales, and can flip the sign of, the TI response; in the dynamic detector the same shifts act through F' and the susceptibility, without a sign rule. The state dependence is the falsifiable handle; the sign reversal is specific to the memoryless model.

3 Methods

3.1 Single-column Jansen–Rit: integration, lock-in, and controls

(All symbols and units are collected in App. A, Table 1.) We integrate the single-column JR model of §2.1 (Eqs. (1), parameters in Table 2) as six first-order ODEs. The TI field enters only the pyramidal output sigmoid, Eq. (1), as $s(t)$; the read-out $v = y_1 - y_2$ is *not* fed the field directly, so any power at Ω in v is produced by the nonlinearity and the recurrent loop, not by trivial pass-through. Unless stated otherwise $f_c = 100$ Hz and $m = 1$.

We integrate with a fixed-step fourth-order Runge–Kutta scheme, $\Delta t = 2 \times 10^{-4}$ s (50 samples per carrier cycle at 100 Hz; reduced to 10^{-4} s for carrier sweeps up to 300 Hz). The kHz-carrier runs of Fig. S9 use $\Delta t = 2 \times 10^{-5}$ s for the carrier sweep to 4 kHz (25 samples per carrier cycle at 2 kHz, 12.5 at 4 kHz) and 1.5×10^{-5} s for the 2 kHz output spectrum. Halving each step changes the reported lock-in amplitudes by less than 1%. The 10 kHz figures quoted in §2.6 are analytic evaluations of Eq. (19), not simulations. For each measurement the system is settled for 10–16 s (longer near the Hopf, where damping is weak) before a 3–20 s measurement window; the resonance sweeps use a 20 s window so the lock-in averages the limit-cycle beating on the entrainment side. Parameter sweeps are vectorized: the state is an $(N, 6)$ array advanced simultaneously for all N parameter combinations, so a full resonance map runs in seconds. The full bifurcation diagram (Fig. 2) is computed by numerical continuation of the fixed-point and limit-cycle branches. We independently locate the operating supercritical Hopf by solving, for each p, the self-consistent scalar equation for the resting fixed point $v = y_1 - y_2$, forming the 6×6 Jacobian numerically and diagonalizing it: the Hopf is the p at which the leading eigenvalue pair crosses the imaginary axis with $\mathrm{Im}\,\lambda_+$ in the alpha band. This gives $p \approx 315$ Hz and $f_0 = \mathrm{Im}\,\lambda_+/2\pi \approx 11.1$ Hz—consistent with the continuation and with the JR bifurcation analysis of [29]. For $p > 315$ the fixed point is a stable focus (resonator); between the saddle-node on an invariant circle (SNIC, $p \approx 114$) and the Hopf the column free-runs on the alpha limit cycle.

The response at the envelope frequency is read out with a software lock-in amplifier referenced to Ω—a standard way to isolate the component of a signal at one chosen frequency: multiply by sine and cosine at Ω and average, so everything *not* at Ω cancels and only the envelope-locked amplitude survives. Concretely we multiply $v(t)$ by quadrature references and integrate over a

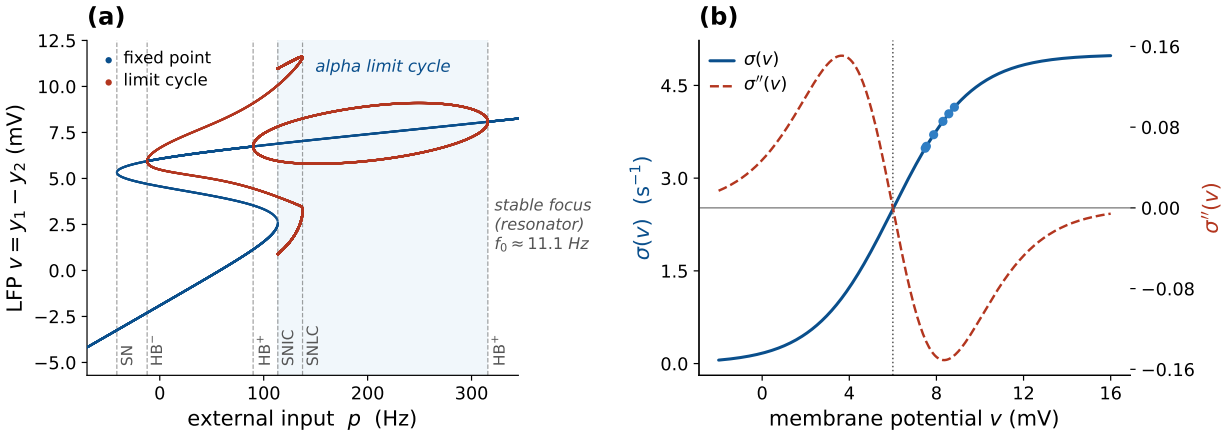


Figure 2: The Jansen–Rit operating regime: bifurcation and sigmoid curvature. *(a)* JR bifurcation diagram from numerical continuation—the LFP $v = y_1 - y_2$ along the fixed-point branch (blue) and the min/max of the limit-cycle branches (red) vs. external input p, with *solid* stable and *faded* unstable. The alpha limit cycle (shaded) is bounded by a saddle-node-on-invariant-circle (SNIC, $p \approx 114$) and the supercritical Hopf ($p \approx 315$, $f_0 \approx 11.1$ Hz); beyond the Hopf the fixed point is a stable focus—the near-critical resonator regime used throughout. The column is a weakly-damped focus near *both* bifurcations, so it resonates on either approach—more slowly near the SNIC. Further codim-1 points are marked (saddle-node SN, Hopf HB$^{\pm}$, saddle-node of limit cycles SNLC). *(b)* The sigmoid $\sigma(v)$ and its curvature $\sigma''(v)$, which sets the demodulation gain and vanishes at the inflection v_0. Dots mark the operating points reached by the closed-loop model for the p values used here.

Hann-windowed measurement window to obtain the in-phase (I) and quadrature (Q) components,

$$I = \frac{2}{\sum_k w_k} \sum_k w_k \, (v_k - \bar{v}) \cos \Omega t_k, \quad Q = \frac{2}{\sum_k w_k} \sum_k w_k \, (v_k - \bar{v}) \sin \Omega t_k, \quad A_\Omega = \sqrt{I^2 + Q^2}, \quad (22)$$

with w a Hann window and $\bar{v} = \sum_k w_k v_k / \sum_k w_k$ the windowed mean. For a pure tone $v = A \cos(\Omega t + \phi)$ the product-to-sum identity gives $I \to A \cos \phi$ and $Q \to -A \sin \phi$, so $A_\Omega \to A$ exactly—the factor $2/\sum_k w_k$ is the normalization that returns the physical amplitude, and A_Ω equals twice the magnitude of the windowed Fourier coefficient at Ω. The one subtlety is that out-of-band components must not leak into the estimate; the dangerous one is the DC level of v ($\bar{v} \sim 8$ mV, far larger than the demodulated signal). Over an exact integer number of Ω-periods $\sum_k w_k \cos \Omega t_k = 0$ and there is no leakage, but in general it is nonzero, so we subtract $\bar{v}$ explicitly; this makes the metric exact for a tone and robust down to the 10^{-8} level needed for the kHz analysis (§2.6). The signed in-phase component I alone recovers $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ including its sign, which we use to verify the curvature law (Fig. 6).

Three auxiliary constructions support the main results. Throughout we contrast the *open-loop detector*—the firing-rate sigmoid alone at a fixed operating point v^*, with no recurrence—against the *closed-loop* column, the full recurrent Jansen–Rit with its feedback. The open-loop detector passes the two-tone field through σ at v^* and applies Eq. (22), isolating the pure demodulation law (Figs. 5, 6, S5). The *bifurcation diagram* records the field-free min/max of v vs. p, each run initialized at its fixed point plus a small kick. The *two-dimensional map* evaluates A_Ω over an (Ω, p) grid. We verify the mechanism three ways: the square law (A_Ω vs. ε on a well-damped focus, expecting a log–log slope of 2); a linearization control (replacing σ in Eq. (1) by its tangent at v^*, which must abolish the Ω response); and a check of σ', σ'' against finite differences. All code is openly available (see *Code and data availability*).

3.2 Laminar two-band model (LaNMM)

We complement the single-column analysis with simulations of the laminar neural mass model (LaNMM) of §2.1. We integrate the full 28-variable column [12, 23], with the field entering exactly

as in the single JR column—as a membrane polarization added inside the sigmoid argument of the target population,

$$\varphi_X(t) = \sigma\big(v_X(t) + s(t)\big), \qquad X \in \{P_1, P_2\}, \tag{23}$$

with $s(t)$ the AM or two-tone field of §2.2 (amplitude ε in mV, beat $\Omega = 2\pi\Delta f$, carrier $\omega_c = 2\pi f_c$) and all three external drives held flat at their baselines (Fig. S2). Coupling the field this way, rather than as a perturbation of an external input *rate*, matters for more than tidiness: a rate cannot go negative, so a rate-coupled AM drive must be clipped at zero once its excursion exceeds the baseline, and a hard clip is a half-wave rectifier—a diode detector placed upstream of the sigmoid, which would inject a line at Δf into the drive before the population nonlinearity ever saw it. A signed membrane polarization carries no such constraint and none is applied. Because the LaNMM and JR sigmoids share the voltage scale $1/\rho \approx 1.8$ mV, ε is directly comparable between the two models. Integration is a vectorized fixed-step RK4 ($\Delta t = 10^{-4}$ s) advancing the whole parameter grid simultaneously, as in the JR sweeps. In place of the lock-in (Eq. (22)) we read the alpha-band (8–12 Hz) power of the P_1 and P_2 membrane potentials as the mean squared magnitude of the Hann-windowed FFT over the band, in arbitrary units, over a post-transient window, a phase-agnostic measure suited to the entrainment regime, where lock-in phase slips would otherwise complicate the read-out. Two sweeps probe the column: an amplitude–beat sweep over $(\varepsilon, \Delta f)$ at fixed carrier (the Arnold tongue), and a carrier–beat sweep over $(f_c, \Delta f)$ at fixed ε, built on the LaNMM module of [12]. The maps (Figs. S1, S7) show the *signed* stimulation-induced change, alpha power minus the unstimulated ($\varepsilon = 0$) baseline, on a zero-centered scale; over most of the P_1-driven grids the change is negative. For the alpha resonance we instead drive the deep P_1 loop directly and sweep its input against Δf, reading the lock-in (Eq. (22)) of v_{P_1} at Δf.

3.3 Exact next-generation mean field (NMM2 PING)

We test the mechanism in the exact next-generation mean field (NMM2, §2.1), following the Rosetta Stone formulation of [14] (which places the Montbrió–Pazó–Roxin reduction [20] on a common footing with the JR and LaNMM models used here), where the firing-rate nonlinearity is the derived MPR dynamics of Eq. (4) rather than a postulated sigmoid. We use a two-population excitatory–inhibitory (PING) motif [21, 22, 14]: each population $\alpha \in \{E, I\}$ is an MPR mean field (firing rate r_α, mean potential v_α) with membrane time τ_α, coupled through second-order synapses,

$$\begin{aligned} \tau_E \dot r_E &= \Delta/(\pi\tau_E) + 2 r_E v_E, \qquad & \tau_E \dot v_E &= v_E^2 + \bar\eta - (\pi\tau_E r_E)^2 + \tau_E C\,(s_E - s_I) + \delta V(t), \\ \tau_I \dot r_I &= \Delta/(\pi\tau_I) + 2 r_I v_I, \qquad & \tau_I \dot v_I &= v_I^2 + \bar\eta - (\pi\tau_I r_I)^2 + \tau_I C\,(s_E - 2 s_I), \end{aligned} \tag{24}$$

with second-order synaptic activations $s_E = \hat K_a[r_E]$ (excitatory, time constant τ_a) and $s_I = \hat K_g[r_I]$ (inhibitory, τ_g), each obeying the same critically-damped operator as the JR synapse, $\tau^2 \ddot s + 2\tau \dot s + s = r$; here C is the global synaptic gain and the weights $(A_{ee}, A_{ei}, A_{ie}, A_{ii}) = (1, 1, 1, 2)$ set the E and I rows—i.e. the E membrane sees $C\,(s_E - s_I)$ and the I membrane $C\,(s_E - 2s_I)$, the only asymmetry being the doubled $I \to I$ weight $A_{ii} = 2$. The common background drive $\bar\eta$—the same in both populations, with no separate input current—is the bifurcation parameter: raising it past a supercritical Hopf at $\bar\eta_H = 1.110$ opens a gamma limit cycle (the gamma counterpart of the JR alpha Hopf). The applied field enters the excitatory membrane current as $\delta V(t) = \lambda E(t)$ as two genuine tones at $f_c \mp \Delta f/2$ (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band); the polarization is expressed in the dimensionless MPR units of v_E, so the carrier is rectified by the exact quadratic v_E^2 term, with no postulated sigmoid. We integrate with fixed-step RK4 ($\Delta t = 0.05$ ms) and read out the lock-in (Eq. (22)) of r_E at Δf over the $(\Delta f, \bar\eta)$ plane. Parameters (Rosetta Stone PING [14]): $\Delta = 1$, global coupling $C = 15$, $\tau_E = 15$, $\tau_I = 7.5$, $\tau_a = 10$, $\tau_g = 2.5$ ms, with $\bar\eta$ swept through the supercritical Hopf at $\bar\eta_H = 1.110$ ($f_0 = 54.0$ Hz), located by bisection on the leading eigenvalue of the mean-field Jacobian.

Two parameters reach this gamma Hopf, and we use both as criticality knobs—just as the external drive p and the coupling C both reach the JR alpha Hopf. The background drive $\bar\eta$ (above)

is swept for the resonance map (Fig. S8); the global coupling $C\,(\equiv J)$ is swept for the J-curve (Fig. S10): we vary J at fixed $\bar{\eta}$ toward the gamma Hopf J^* (located by bisection on the leading Jacobian eigenvalue) and read the lock-in peak of r_E over Δf. Unlike the JR J-curve, *no operating-point pinning is needed*: at this carrier the generated difference-frequency source is operating-point independent to about 1% (§2.3), so sweeping J changes the damping γ and the modal projections but not the source, and the gain is read as the response per unit source.

3.4 QIF spiking network (microscale of NMM2)

NMM2 is the exact mean field of an all-to-all network of QIF neurons [20, 14, 22], so we integrate that network directly to test the prediction at the spike level (§4.8): $N = 8000$ QIF neurons per population, parameters identical to the PING mean field (Eq. (24)), excitabilities $\eta_j \sim \text{Lorentzian}(\bar{\eta}, \Delta)$ sampled by deterministic quantiles, second-order synapses driven by the empirical population rates, and a genuine two-tone field entering the excitatory membranes. Spiking and threshold dynamics are where the AM surrogate and the physical waveform cannot be assumed equivalent beyond perturbation theory, so this run uses two tones throughout. The bifurcation parameter is the common background drive $\bar{\eta}$, with the gamma Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz) from the mean-field Jacobian; the spiking network shows the same threshold and an autonomous $f_0 \approx 53$ Hz, with no finite-size offset to explain. The carrier is $f_c = 300$ Hz and the field amplitude $A_f = 35$. Because this is a *gamma* circuit, the TI beat Δf is set in the gamma band: $\Delta f = 55$ Hz *near* the forced gamma resonance ($\bar{\eta} = 0.85$, below the Hopf) and $\Delta f = 42$ Hz in the oscillatory regime ($\bar{\eta} = 1.5$, above the Hopf)—a detuning of $\approx$11 Hz below the autonomous $f_0 \approx 53$ Hz that places the beat outside the 1:1 Arnold tongue, so the above-Hopf response is gating, not entrainment (§F). The network is integrated with Euler steps of 0.0025 ms, settled for 800 ms (over four relaxation times at $\bar{\eta} = 0.85$, where the mean-field relaxation time is 176 ms) and measured over an integer number of beat cycles, 44 at 55 Hz and 34 at 42 Hz (800 and 810 ms), so that the finite-window bias of phase statistics vanishes. Envelope-locked timing is quantified by the quadrature lock-in A_Ω of $r_E(t)$ at Δf, reported as the modulation depth $A_\Omega/\langle r_E \rangle$ (a stable measure; the off→on lock-in ratio would divide by the field-free floor), and per neuron by the pairwise phase consistency of each neuron's spike phases at Δf [30], which is unbiased in the spike count, over neurons with at least ten spikes in the window. Halving the time step changes the lock-in and the pooled vector strength by under 1%; doubling the spike-detection threshold changes the lock-in by about 1% and the mean rate by 2%, but the pooled vector strength in the forced regime by about 45%, so phase concentration, unlike rate modulation, is sensitive to how spike times are defined.

3.5 Coupling sweeps, timing, and the tACS control (Jansen–Rit)

The coupling sweeps of Figs. 9 and S11, and the transcranial alternating-current stimulation (tACS) control of Fig. S3, use the single JR column with the operating point v^* pinned: for each global connectivity C we solve the field-free fixed-point relation analytically for the external drive $p(C)$ that holds v^* (hence $\sigma''(v^*)$) fixed, vary $C\,(= J)$ toward the alpha Hopf, and read γ, ω_0 from the 6×6 Jacobian. The field is the AM surrogate of Eq. (11) for TI (Fig. 9; AC lock-in vs DC mean rate for Fig. S11) or a direct in-band sinusoid $s(t) = \varepsilon \cos \Omega t$ ($\varepsilon = 0.1$ mV) for the tACS control (Fig. S3), each read out by the lock-in of v at Δf.

3.6 Vocabulary: forced response, entrainment, and gating

"Entrainment" is used in several senses, so we fix three terms. *Forced response* is the driven, non-autonomous case below the Hopf, quantified by the lock-in amplitude at the beat Δf (Eq. (22)). *Entrainment* is reserved for its dynamical-systems meaning, frequency capture of a *self-sustained* oscillator [31]; in the maps we screen for it with the dominance criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz on the dominant output frequency, a necessary but not sufficient condition for capture. *Gating* is amplitude modulation of a persisting rhythm without frequency capture [32, 33]. Capture and

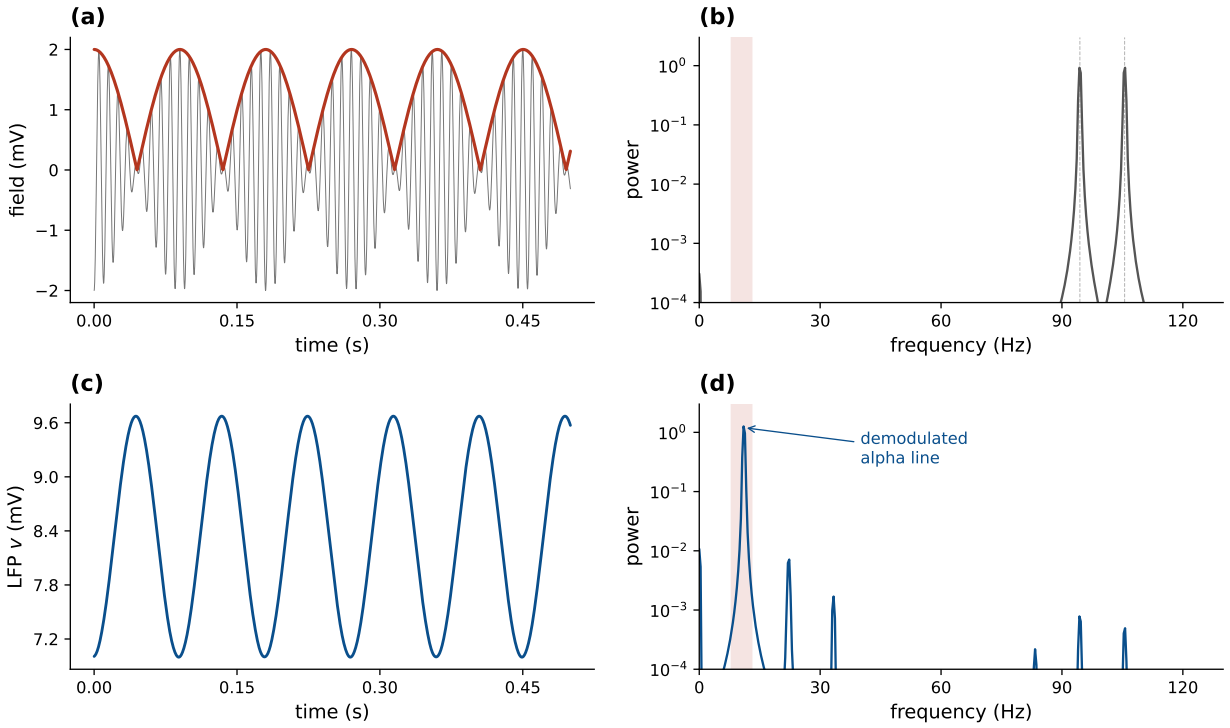


Figure 3: An alpha rhythm is synthesized from a two-tone TI input (Jansen–Rit below Hopf). The field is the physical TI waveform, two equal tones at $f_{1,2} = f_c \mp \Delta f/2$, not the amplitude-modulated surrogate. *(a)* The field $s(t)$ and its envelope $A(t) = 2\varepsilon|\cos(\Omega t/2)|$—a rectified cosine, not a sinusoid; it is A^2 that is sinusoidal at Ω (§2.2). *(b)* The input amplitude spectrum has lines only at f_1 and f_2 (94.45 and 105.55 Hz, $\Delta f = 11.1$ Hz) and no third line at the carrier; in the alpha band (shaded) it is 1.6×10^{-8} of the tone amplitude, finite-window leakage rather than physical power. *(c)* The output $v = y_1 - y_2$ oscillates at the envelope frequency. *(d)* The output spectrum has a strong peak at the beat, 11.1 Hz, synthesized by the sigmoid and absent from the input.

gating are dynamically different, and the same word would otherwise cover both; Appendix F gives the metrics and the Arnold-tongue protocol.

4 Results

Two rungs of the modeling ladder carry the evidence: the single Jansen–Rit column (closed form, memoryless detector) and the exact next-generation mean field (NMM2, the nonlinearity derived rather than postulated) with its spiking network. Each shows the same pair, quadratic difference-frequency generation followed by near-Hopf amplification, with different detector coefficients. Two mesoscale signatures follow: the network's own coupling sets the gain at fixed generated drive (the J-curve, §4.6), and the resonance amplifies the envelope-locked modulation far more than the mean rate (§4.7), which the QIF network tests at the spike level. The laminar two-band column, at carriers of 25–80 Hz and drives up to 3 mV, is an extension (App. E).

4.1 The Jansen–Rit sigmoid demodulates: an alpha line from a carrier-only input

The system is oriented by Fig. 2: a clean supercritical Hopf at $p \approx 315$ with $f_0 \approx 11.1$ Hz, and a sigmoid whose curvature peaks well off the inflection, so the closed-loop operating points sit where σ'' is large. The core result is Fig. 3: an input with no spectral power in the alpha band (lines only at f_c and $f_c \pm \Delta f$) produces a clean ∼11 Hz output oscillation locked to the envelope, with a strong alpha peak in the output spectrum absent from the input.

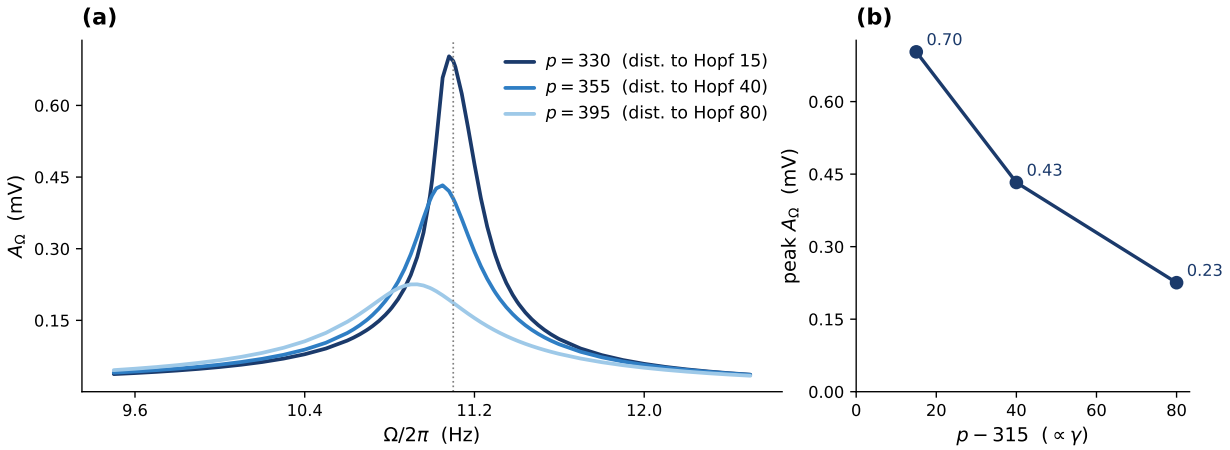


Figure 4: The demodulated beat drives a resonance that sharpens toward the Hopf (Jansen–Rit). Fixed-point regime ($p > 315$, stable focus): lock-in response at Ω vs. envelope frequency ($\varepsilon = 0.3$ mV). *(a)* The demodulated drive forces a Lorentzian resonance at $f_0 \approx 11.1$ Hz whose peak redshifts slightly as the damping grows ($\omega_{\text{peak}} = \sqrt{\omega_0^2 - 2\gamma^2}$; $11.08 \to 10.93$ Hz across the three curves); the response carries no power at Ω in the absence of drive, so this is genuine forced gain. *(b)* The peak height grows as the Hopf is approached ($p \to 315$, distance to Hopf $\to 0$). Three points cannot establish a power law; the $1/\gamma$ scaling is quantified over a decade, together with its saturation, in Fig. 9(c). The complementary above-Hopf regime, where the demodulated drive *entrains* the autonomous alpha within a 1:1 Arnold tongue, is characterized with the appropriate observable (output frequency f_{out}, not lock-in amplitude) in Fig. S4.

4.2 Near the Hopf, the demodulated beat drives a sharp resonance

The resonance is quantified in Fig. 4. In the fixed-point regime ($p > 315$, a stable focus) the response is a Lorentzian centered at f_0 whose height grows monotonically as the Hopf is approached ($0.23 \to 0.43 \to 0.70$ mV; Fig. 4), i.e. gain $\propto 1/\gamma$. This is genuine forced gain: with no drive the focus is silent at Ω, so all power at Ω is synthesized by the demodulating nonlinearity and amplified by the recurrent loop. Above the Hopf ($p < 315$) the column free-runs at f_0 and the demodulated drive instead phase-locks that autonomous alpha within a 1:1 Arnold tongue; because lock-in amplitude at Ω then conflates the driven response with the free-running cycle, we characterize that regime with the output frequency f_{out} rather than the lock-in amplitude (Fig. S4).

4.3 Carrier independence and the kHz roll-off

At fixed post-coupling polarization the demodulated response is flat across the carrier band, for both the open-loop detector and the closed-loop column near the Hopf (Fig. 5). As a function of the *applied* field, by contrast, the membrane low-pass makes the response roll off as $1/f_c^2$ above the corner (Fig. S5): passive somatic demodulation is suppressed four to six orders of magnitude at kHz, whereas a fast element ($\tau \approx 0.2$ ms) retains an appreciable response into the kHz band.

To pre-empt the objection that the demonstration runs only at 100 Hz, we drive the full recurrent column with a genuine kHz field, passing it through an explicit membrane low-pass (an extra first-order state, time constant τ) for each coupling route before it reaches the sigmoid. The column demodulates and resonates at the recovered alpha well into the kHz band when the nonlinearity sits at a fast element ($\tau \approx 0.2$ ms), exactly as Eq. (19) predicts (Fig. S9a). Driving the fast-element route with two *genuine* tones at $f_{1,2} = 1994.54$ and 2005.46 Hz, centered near 2 kHz and beating at 10.91 Hz, the input carries 2.2×10^{-13} of its spectral amplitude in the 8–14 Hz band, i.e. none, while the output carries a line at the beat (Fig. S6), with the attenuation set by the analytic $|H_m|^2 = 0.137$ at 2 kHz for $\tau = 0.2$ ms. The kHz claim therefore rests on the physical waveform, not on an extrapolation from the AM surrogate. That the bare unmodulated kHz carrier is itself weakly effective—the premise of TI's focality—is consistent with the limited sensitivity of hippocampal network oscillations to unmodulated kHz fields [34].

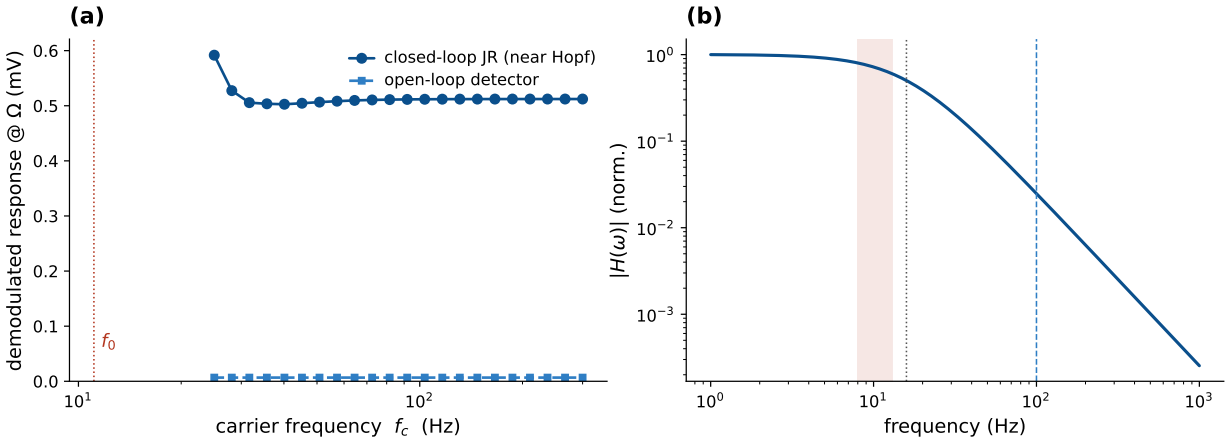


Figure 5: The demodulated response is carrier-independent at fixed polarization (Jansen–Rit). *(a)* The demodulated response is flat across carrier frequency $f_c = 25$–300 Hz, both for the open-loop detector (exactly, by Eq. (9)) and for the closed-loop JR near the Hopf (which adds $\sim 60\times$ resonant gain); deviations appear only as f_c approaches the natural band. *(b)* The second-order-synapse low-pass $|H_s(\omega)|/|H_s(0)|$ passes the envelope (~ 0.7) and suppresses the 100 Hz carrier ($\sim 2.5\%$). Dotted line: synaptic corner $a/2\pi \approx 16$ Hz; shaded band: envelope (8–13 Hz); dashed line: 100 Hz carrier.

4.4 The nonlinearity is the detector: the σ'' curvature law and its controls

The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ across the operating range, vanishing and reversing sign at the inflection v_0 (Fig. 6)—demodulation is the sigmoid-curvature term of Eq. (9), and the closed-loop operating points sit where σ'' is large and negative.

Two controls test the proposed source of the beat component (Fig. 6b,c): the response scales as ε^2 at small drive (measured log–log slope 1.99 against the predicted 2), and replacing the field-receiving sigmoid by its linearization collapses the response by $634\times$ ($1.34 \to 0.0021$ mV)—the nonlinearity is the detector, and its curvature is the gain. A third panel checks the surrogate at one operating point: for the open-loop detector at $v^* = 8$ mV the two-tone and AM responses at Ω agree to within 10^{-4} at $\varepsilon = 0.02$ mV and to 1.2% at 1 mV (Fig. 6d). On that basis the dense perturbative sweeps below, which read only the Ω component, use the surrogate; the comparison does not test the recurrent response near the Hopf, where the two waveforms' different DC terms could shift the operating state.

4.5 Exact mean-field confirmation: the NMM2 PING gamma generator

The JR column and the LaNMM both use the heuristic firing-rate sigmoid (NMM1). As a third, independent test we move to the exact next-generation mean field (NMM2, Eq. (4)), where the demodulating nonlinearity is the *derived* quadratic v^2 term rather than a postulated curve. We drive the two-population E–I PING motif of Eq. (24) (Methods)—a gamma generator that loses stability through a supercritical Hopf (at $\bar{\eta}_H = 1.110$, $f_0 = 54.0$ Hz), the gamma counterpart of the JR alpha Hopf—with the two-tone field entering the excitatory current, so the carrier is rectified by the exact v_E^2 term.

Driving with a two-tone TI field (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band) and locking in the excitatory rate r_E at Δf reproduces the whole picture (Figs. 8, S8). Below the Hopf the response is a forced resonance at the gamma frequency whose height grows as the bifurcation is approached—the $1/\gamma$ gain of Eq. (18); above it the demodulated drive entrains the autonomous gamma. The two-dimensional map traces a ridge at $\Delta f \approx f_0$ that intensifies through the Hopf. Unlike the static-sigmoid models, the resonant frequency itself *shifts with the operating point*—the ridge is diagonal, not vertical—a direct signature of the dynamic, state-dependent transfer of the exact theory [22]. That the beat is generated and resonantly amplified with the

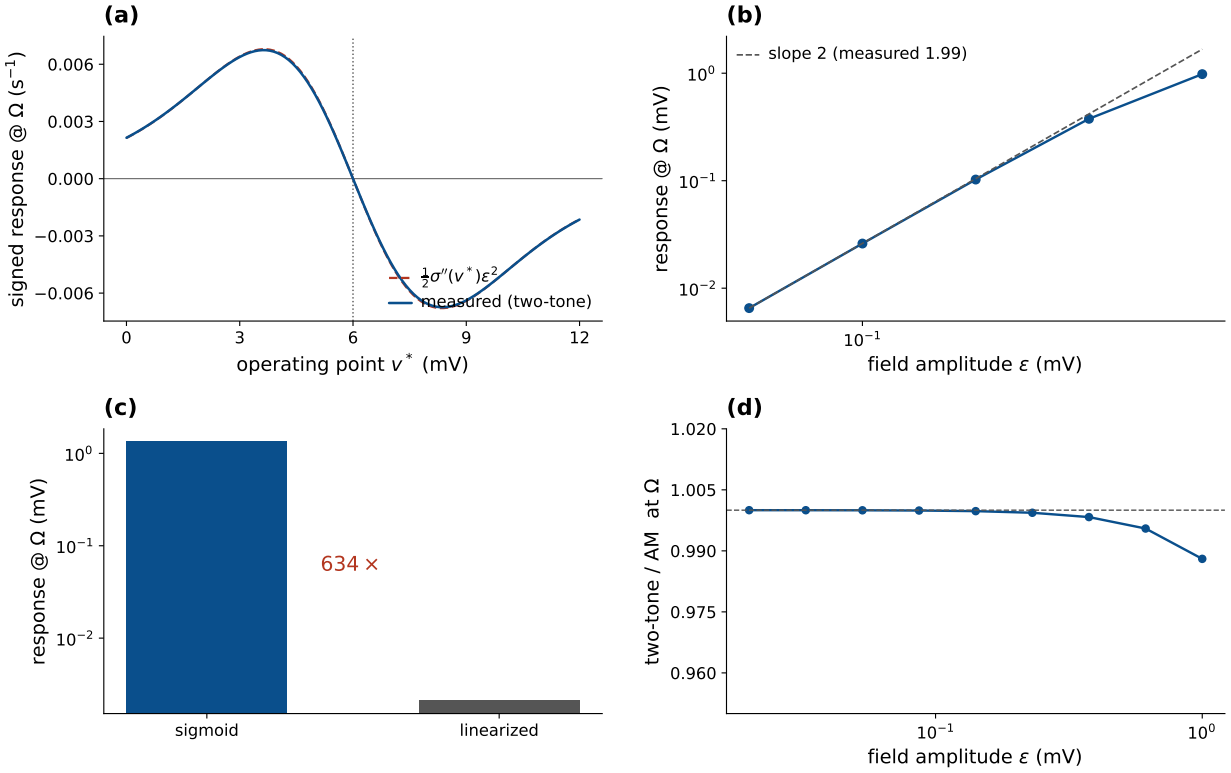


Figure 6: The curvature law, the square law, and the surrogate control (Jansen–Rit, genuine two tones). *(a)* The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2$ across the operating range (maximum deviation 0.9%), vanishing at the inflection v_0 and reversing sign across it. The mechanism is sign-agnostic: σ'' sets sign and phase while $|\sigma''|$ sets gain, so the resonance picture is unchanged up to an overall sign in the $\sigma'' > 0$ lobe, which this sweep spans. The sign flip is itself a prediction (§4.9). *(b)* The closed-loop response scales as ε^2, measured slope 1.99. *(c)* Replacing the field-receiving sigmoid by its tangent collapses the response 634×. *(d)* Surrogate control: the two-tone/AM ratio at Ω approaches 1 as $\varepsilon \to 0$ (1.0000 at 0.02 mV, 0.988 at 1 mV), licensing the AM surrogate for the dense sweeps of Fig. 9, which read only that component, and for nothing else.

nonlinearity *derived* from QIF spiking dynamics, rather than assumed, shows the receiver behavior is not an artifact of the heuristic sigmoid. The detector coefficient, however, is not the sigmoid's: at this 300 Hz carrier the population is on the fast side of its crossover, and the generated source is the v_E^2 mixing of the carrier-frequency voltage excursion, operating-point independent to about 1% (§2.3, Fig. 7).

4.6 What the mesoscale adds: coupling sets the gain at fixed detection

The resonance sweeps above drove the column toward the Hopf with the external input p, but p also shifts the operating point, entangling detection (σ'') with amplification. The sharper, mechanistic statement is that the *network coupling* sets the gain on its own. In the JR column we vary the global connectivity C—the coupling strength J—and, for each C, solve analytically for the drive $p(C)$ that *pins* the operating point v^*, so the detection gain $\sigma''(v^*)$ and the open-loop detector amplitude $A_{\text{open}} = \frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ are held exactly constant. As $C \to C^*$ (the Hopf, $C^* \approx 137$) the damping $\gamma \to 0$, the demodulated response grows ∼30× and the envelope resonance sharpens (Fig. 9a,b)—entirely a network effect, since detection is frozen and only γ changes. The recovered gain A_Ω/A_{open} tracks $1/\gamma$ before saturating near criticality (Fig. 9c); driving the same column to the Hopf through the input p rather than the coupling C yields the same gain–γ relation, so the amplification depends only on the distance to criticality, not on which parameter delivers it. All these sweeps stay on the stable-focus side ($\gamma > 0$), approaching the supercritical Hopf from below: the $1/\gamma$ growth is the forced-resonance susceptibility of a damped network, not entrainment of

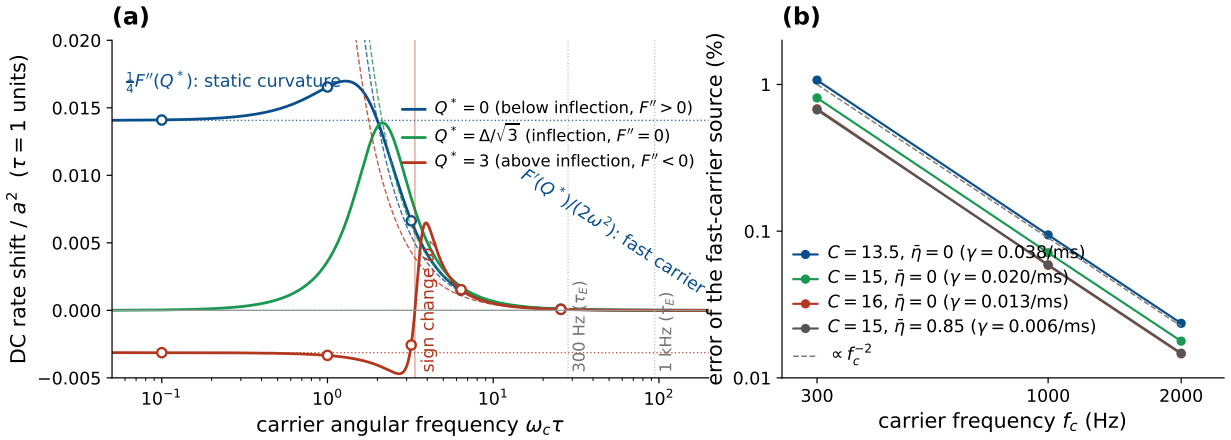


Figure 7: Static curvature at slow carriers, slope at fast ones. *(a)* Second-order DC rate shift of an uncoupled MPR population ($\Delta = 1$, $\tau = 1$) under a current carrier, Eq. (12), for operating points below, at and above the inflection of $F(Q)$. Dotted: the static limit $\frac{1}{4}F''$; dashed: the fast-carrier limit $F'/(2\omega^2)$; circles: RK4 integrations at $a = 0.05$ averaged over whole carrier periods (relative error below 10^{-3}). Above the inflection the sign changes at ω_z. The top axis converts $\omega\tau$ to a carrier frequency for $\tau = \tau_E = 15$ ms. *(b)* PING of Eq. (24): relative error of the fast-carrier difference-frequency source against the exact second-order response at four stable operating points (γ the damping of the gamma mode), two tones centered at 300, 1000 and 2000 Hz.

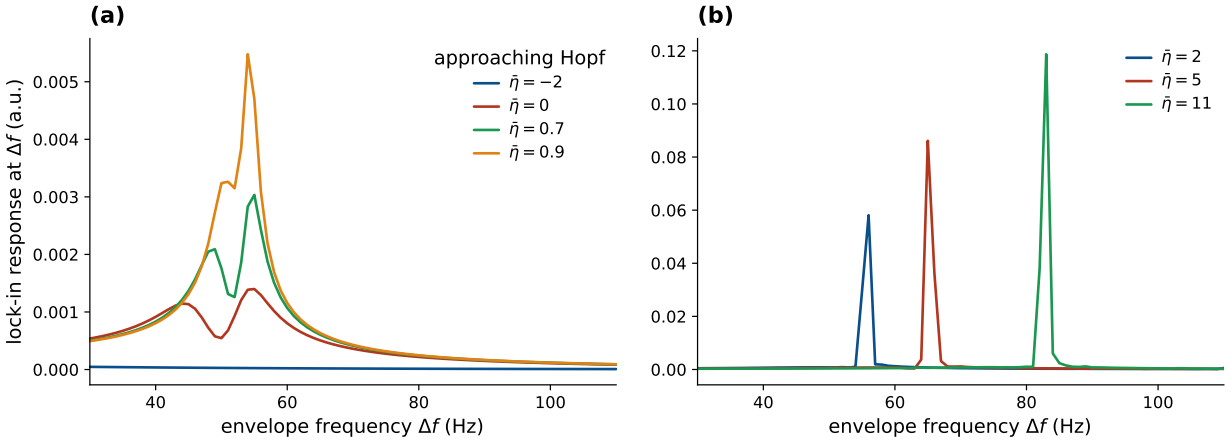


Figure 8: NMM2 PING gamma resonance (exact mean field, genuine two tones). Lock-in response of the excitatory rate r_E at the envelope frequency Δf under a two-tone TI field (carrier 300 Hz). *(a)* (stable focus, $\bar{\eta} < \bar{\eta}_H = 1.110$) forced resonance at the gamma frequency ($f_0 \approx 54$ Hz), the peak growing $0.0014 \to 0.0030 \to 0.0055$ as the Hopf is approached ($1/\gamma$ gain). *(b)* (limit cycle, $\bar{\eta} > \bar{\eta}_H$) entrainment of the autonomous gamma, the peak shifting upward with $\bar{\eta}$ ($56 \to 65 \to 83$ Hz). The input carries no power at Δf; the response is synthesized by the exact quadratic v^2 nonlinearity, with no sigmoid anywhere in the model. Matched AM and two-tone drives agree at Ω to 0.07% well below the Hopf and to 4% at $\bar{\eta} = 0.9$, consistent with the JR surrogate control (Fig. 6d).

an autonomous cycle. On the other side of the bifurcation the column oscillates on its own, and the demodulated drive instead phase-locks that limit cycle within an Arnold tongue (Fig. S4; also the entrainment side of Fig. 4 and the LaNMM and NMM2 maps)—1:1 frequency capture over a band that widens with drive, quantified by the locking range rather than a diverging $1/\gamma$ gain, the two coinciding only at onset. Because the drive is weak, on the autonomous side it biases phase (timing) more than amplitude, consistent with the timing-not-rate signature (§4.7).

The next-generation theory gives the same result without pinning. In the NMM2 PING the coupling C is the *literal* mean inter-neuron (QIF) synaptic weight J, and at this carrier the generated source is operating-point independent to about 1% (§2.3), so detection needs no pinning.

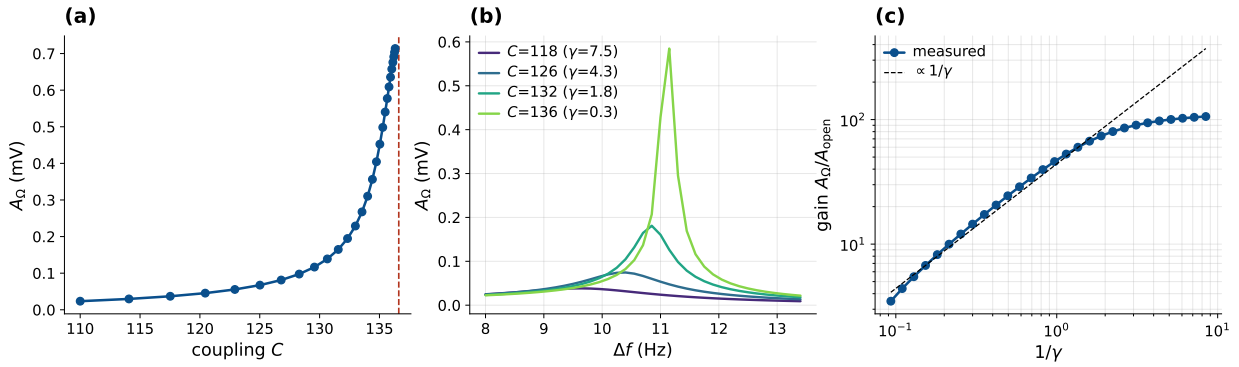


Figure 9: Coupling controls the amplification at fixed detection (Jansen–Rit). The operating point v^* is pinned by co-solving $p(C)$, so $\sigma''(v^*) = -0.151$ and A_{open} are held constant throughout. *(a)* The demodulated A_Ω grows as the coupling $C\,(= J)$ approaches the Hopf C^* from the stable-focus side ($\gamma > 0$). *(b)* The envelope resonance sharpens and grows as $C \to C^*$. *(c)* The network gain A_Ω/A_{open} follows $1/\gamma$ (dashed) and saturates near the bifurcation.

Sweeping J at fixed background drive $\bar{\eta}$ toward the gamma Hopf J^*, the demodulated peak of r_E grows as $1/\gamma$ and saturates at criticality (Fig. S10): the heuristic JR result, reproduced with the coupling *and* the rectifier both derived rather than assumed, and a direct counterpart of the coupling-controlled weak-field sensitivity of QIF networks [22]. The two columns realize one band-agnostic law in their two native bands—alpha (JR) and gamma (NMM2): the single-cell nonlinearity detects the envelope (coarse-grained into σ'', or the exact v^2), while the sharp, tunable gain is the network's—a resonant amplification set by proximity to the Hopf and adjustable through the coupling.

4.7 The resonance amplifies the envelope-locked modulation more than the mean rate

The demodulator feeds the network a constant term and a line at the beat Ω (Eq. (8)), and the transfer $G(\Omega)$ of Eq. (18) treats them differently: $G(\omega_0) \propto 1/(2\gamma\omega_0)$ diverges as the Hopf is approached, while $G(0) \propto 1/\omega_0^2$ stays finite. Pinning v^* to hold the local detector curvature fixed, as in Fig. 9, and driving the column toward the Hopf through the coupling, we read both quantities from the same observable, the pyramidal rate $r_P = \sigma(v + s)$ with the field in its argument. The beat-locked modulation of r_P at $\Delta f = f_0$ grows 13.5× (26 to 346 mHz), tracking $1/\gamma$ before saturating near criticality; the shift of its mean grows too, 5.8× (−1.3 to −7.7 mHz), through rectification of the growing oscillation, but stays below 0.2% of the 3.9 Hz baseline (Fig. S11). Modulation exceeds mean shift by 19× far from the Hopf and by 45–66× near it. The network amplifies the envelope-locked modulation far more than the mean rate, so a small mean-rate change is the expected signature even where TI is acting, the mesoscopic counterpart of the observation that TI alters spike timing more than firing rate in the primate brain [6]. Finite is not zero: the mean does move, and the comparison must use commensurate measures, as here.

4.8 Microscale confirmation: TI realigns spike timing in the QIF network

The mesoscale prediction has a direct microscale test, since NMM2 is the *exact* mean field of an all-to-all network of quadratic integrate-and-fire (QIF) neurons with Lorentzian-distributed excitabilities [20, 14, 22]. Integrating that spiking network ($N = 8000$ neurons per population, parameters identical to the PING mean field; Methods) shows TI bunching spike *times* into the beat cycle with a smaller change in the mean rate, on both sides of the gamma Hopf (Fig. 10). Below the Hopf (forced) the field-free network fires asynchronously at a flat rate; with TI on, spikes condense into envelope-locked bands (Fig. 10a,b). Above the Hopf (oscillatory) the free-running gamma is *gated* by the TI beat—forced cross-frequency modulation (phase–amplitude coupling, PAC), not frequency entrainment, since the ≈11 Hz detuning lies outside the 1:1 tongue: its bursts

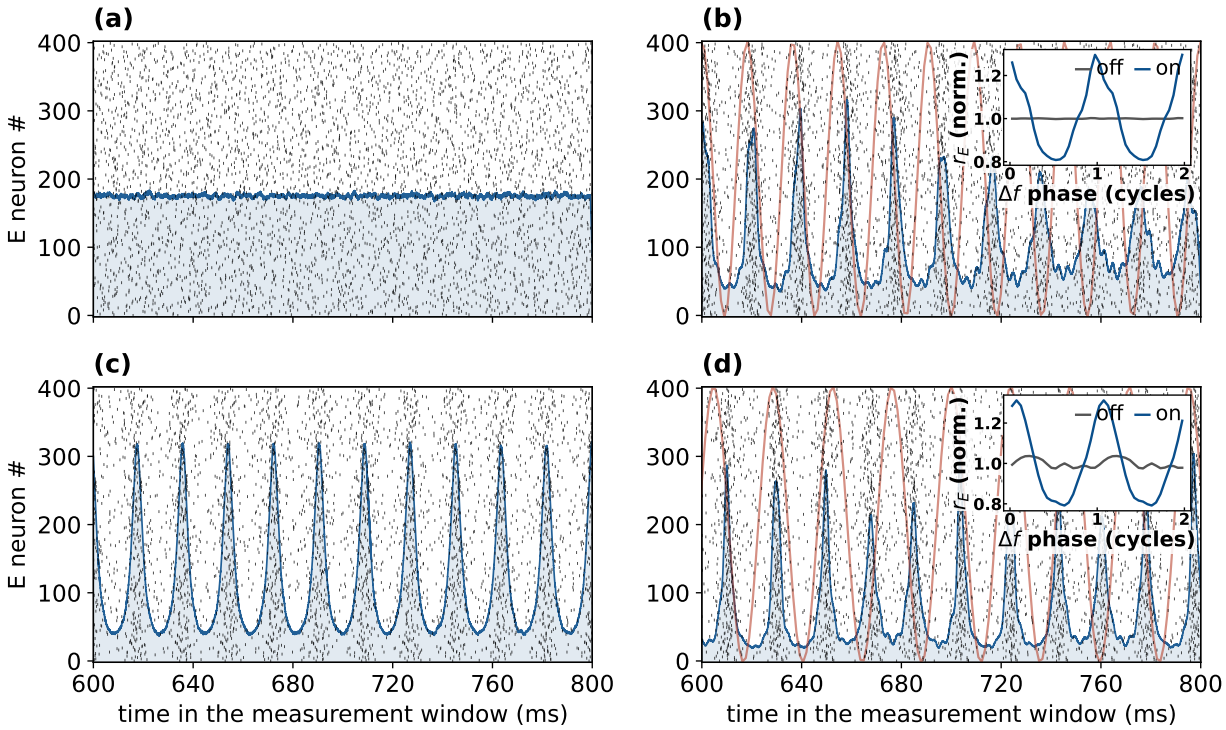


Figure 10: QIF spiking network: TI bunches spikes into the beat cycle with a smaller change in mean rate. $N = 8000$ excitatory and inhibitory quadratic integrate-and-fire neurons whose exact mean field is the NMM2 PING circuit (Eq. (24)). The circuit's intrinsic rhythm is gamma (∼53 Hz), so the beat is set in the gamma range (carrier $f_c = 300$ Hz, genuine two tones). Black ticks are E spikes (400 of 8000), blue the population rate, red the TI envelope; the bifurcation parameter is the common drive $\bar{\eta}$ (gamma Hopf at ≈1.1). *(a,b)* Forced ($\bar{\eta} = 0.85$), TI off then on at $\Delta f = 55$ Hz: asynchronous firing becomes envelope-locked bunching. *(c,d)* Oscillatory ($\bar{\eta} = 1.5$), TI off then on with the beat detuned to 42 Hz: the free-running gamma persists but gains a Δf-locked component. Because the ≈11 Hz detuning places the beat outside the 1:1 tongue, this is forced cross-frequency *gating*, not frequency capture (contrast Fig. S4); the detuning is what certifies the locked component as a driven effect. The last 200 ms of the 800 ms measurement window are shown. *Insets* in *(b,d)*: the population rate folded on the beat phase over the whole window (44 and 34 cycles), each normalized to its own mean (off gray, on blue); with that many cycles the near-commensurate 53 Hz gamma averages out of the 42 Hz fold in *(d)*. Both are flat with TI off.

group into the $\Delta f = 42$ Hz envelope while the intrinsic ∼53 Hz gamma persists (Fig. 10c,d). Because the two rhythms coexist in a nonlinear medium, the QIF nonlinearity mixes them into intermodulation sidebands in the population rate—most prominently the direct Δf line and the sum component near 95 Hz (53+42), with the difference tone near 11 Hz (the detuning itself); this is the spiking-network fingerprint of the same nonlinear frequency mixing that operates at the macroscale, and it is the sideband signature of *gating* rather than the frequency pulling of entrainment. The spiking network reproduces the exact NMM2 mean field's gamma spectrum (Fig. S12a). Across both regimes the mean rate rises by 8–10%, while the Δf modulation depth of the population rate, $A_\Omega/\langle r_E \rangle$, rises from below 0.003 to 0.29 (forced) and 0.23 (gated) (Fig. S12c), measured over 44 and 34 whole beat cycles after 800 ms of settling (Methods). Timing modulation therefore exceeds the relative mean-rate change by a factor of about three at the population level. Per neuron the locking is weak: the pairwise phase consistency of single-neuron spike phases at Δf rises from its field-free control (median −0.02) to a median of 0.003–0.005, and the pooled vector strength from below 0.02 to 0.12–0.13. The population's beat-locked component is built from many weakly biased cells, not from strongly locked ones. This is the spiking-level counterpart of §4.7 and a qualitative analog of the in-vivo finding that TI alters spike timing more than firing rate [6].

4.9 Falsifiable predictions and tests

The mechanism makes four predictions, each with an experimental handle.

1. **Carrier independence at fixed polarization.** Once the post-coupling polarization $\varepsilon = \lambda_f E_0$ is matched, the leading quadratic beat component no longer depends on f_c, provided the carrier lies outside the responsive bands of both detector and network; frequency dependence reappears through the effective coupling $\lambda_f = \lambda_0 |H_c(f_c)|$, or when a fast cellular or network stage is itself resonant (§E). The *applied*-field threshold still rises with f_c through the membrane roll-off (Eq. (19))—as single-neuron accounts also predict [4, 2]. *Test:* hold the in-zone polarization fixed (raising E_0 to compensate the measured λ_f) while sweeping f_c; the network signature is a flat envelope response. At fixed applied field the same sweep reads the entry point: the roll-off should sit in the kHz range (AIS access, Na_v gating), not at the ∼10 Hz somatic corner (§2.6). Existing AM-versus-tACS comparisons match the extracellular modulation rather than ε [6], so this is the correction they lack.

2. **State- and carrier-dependent response.** The generated beat depends on the operating point through the detector coefficient, and the two modeled detectors disagree on how: the memoryless sigmoid follows $\sigma''(v^*)$ and reverses sign across the inflection, where PEIX (Eq. (21)) peaks; the dynamic mean field at a fast carrier follows F' and the network susceptibility, without a sign rule (§2.3). *Test:* shift v^* with tonic drive, neuromodulation, or task state at a fixed beat, and sweep the carrier at a fixed beat. A sign reversal across the operating point would select the memoryless description, a monotonic change the dynamic one.

3. **Transduction class.** A weak smooth nonlinearity transduces A^2 and an envelope follower transduces A (§2.4), and the two are separable at a single matched target. The former gives $Y_\Omega \propto \varepsilon^2$ with a second-harmonic ratio vanishing as $\varepsilon \to 0$; the latter gives $Y_\Omega \propto \varepsilon$ with the ratio pinned at 1/5. *Test:* sweep field amplitude off resonance (so that both lines sit on the flat part of G) and read the log–log slope together with $Y_{2\Omega}/Y_\Omega$; on resonance the network suppresses the second harmonic by $\sim 3\mathcal{Q}$ and the test must be corrected by an independently measured $G(2\Omega)/G(\Omega)$.

4. **Frequency selectivity and state-dependence.** The response is largest when Δf matches a region's resonant rhythm and is amplified as $1/\gamma$ near a bifurcation, so manipulations that move the operating point or the distance to the Hopf—psychedelics, PV-interneuron dysfunction, attention, arousal—should scale, and can abolish or sign-reverse, TI efficacy [12]. *Test:* sweep Δf through a region's intrinsic band across brain states and track the resonant peak and its width.

Table 2 confronts these predictions—together with two corollaries the mechanism already shares with published observations—against existing evidence. Two are qualitatively consistent with published observations: TI alters spike timing more than firing rate in vivo [6] (our $G(\omega_0) \gg G(0)$ asymmetry, §4.7–4.8), and most pyramidal TI responses vanish when synaptic transmission is blocked [7] (a network read-out, which tests network involvement rather than excluding a surviving cellular detector). A quantitative reanalysis of these datasets against the model is a natural next step, and §5 specifies the amplitude-exponent fit it would require.

5 Discussion

The models show how neural populations can act as tuned receivers of TI inputs. Nonlinear mixing generates a component at the difference frequency, and recurrent dynamics selectively amplify it near a network resonance. Generation and amplification are distinct contributions: a large network response does not, by itself, imply efficient detection. This complements rather than competes with the single-neuron ion-channel account [2]: both are rectify-then-filter demodulation, ours is agnostic about which fast cellular nonlinearity supplies the mixing, and it inherits the

Prediction (model basis)	Existing evidence	Sharp test
Transduction class: $Y_\Omega \propto \varepsilon^2$ and vanishing $Y_{2\Omega}/Y_\Omega$ (square law) vs. $\propto \varepsilon$ and 1/5 (envelope following)	— (untested; existing comparisons match extracellular field, not ε)	Amplitude sweep off resonance; transfer-corrected harmonic ratio
Timing-dominated response: the envelope-locked modulation of the rate grows toward the Hopf faster than its mean ($G(\omega_0) \gg G(0)$)	*Qualitatively consistent*: TI alters spike timing more than firing rate, primate [6]	Lock-in of spike timing at Δf vs. the mean-rate change, in the same effect measure
Network read-out (population nonlinearity + synaptic recurrence)	*Qualitatively consistent*: most pyramidal TI responses vanish under synaptic block [7]	Synaptic block should abolish the demodulated response
Carrier independence at fixed polarization (no ω_c in A_Ω)	TI thresholds inherit the carrier's f_c-dependence [4, 5]; bare kHz weakly effective [34]	Fix in-zone polarization, sweep $f_c \to$ flat envelope response; at fixed field, a kHz-range (not 10 Hz) roll-off
State- and carrier-dependent response (the detector coefficient is model-dependent)	Cell-type-specific TI responses (PV vs. pyramidal) [7]	Shift v^* by state at fixed beat; sweep f_c at fixed beat
Frequency selectivity, sharpened $\sim 1/\gamma$ near criticality	— (a sharp new prediction; largely untested)	Sweep Δf through a region's band across brain states

Table 2: The mechanism's predictions confronted with existing in-vivo / in-vitro evidence. Entries cite documented qualitative findings; quantitative reanalysis is left to future work.

brain's own resonances, so a region's tuning is set by its intrinsic dynamics as a radio is tuned by its tank.

The detector's dynamics matter. The static sigmoid and the dynamic QIF mean field both support quadratic mixing in the weak-response regime, but their mixing coefficients and operating-point dependence differ (§2.3). A steady-state firing-rate curve therefore does not, on its own, determine the response to a fast carrier. The common mechanism is nonlinear mixing followed by network tuning, not a universal sigmoid-curvature law. The single cell need only supply a fast nonlinearity that samples the carrier; in the exact mean field that is the spike-generating quadratic acting on a voltage excursion of order a/ω.

Where the carrier-sampling nonlinearity may live. The literature points to three non-exclusive loci. (i) *Axonal Na^+ channels* at nodes and the AIS, where fast voltage-gated Na^+ rectifies because activation outpaces inactivation [2, 35] and inactivation-mediated block passes the slow envelope while removing the carrier [4]. (ii) *Presynaptic terminals*, which polarize ∼2–10× more than somata [36, 37, 38] and convert depolarization into release through a steeply nonlinear Ca^{2+} relation. Sensitivity is not speed, however: for an isopotential compartment $\tau_m = R_m C_m$ with both quantities per unit area, so the time constant does not shrink with size—a small bouton has small capacitance and correspondingly large input resistance. A terminal therefore tracks a kHz carrier no better than a soma, and the short axonal time constants come from high-conductance nodal membrane rather than from small size. Terminals are the *sensitive* locus, not the fast one, and the carrier sampling is supplied by (i). (iii) *Threshold spiking*, where a weak bias near threshold is rectified through stochastic resonance [28, 27]. Each is a candidate; the simulations impose an access filter and a nonlinearity, they do not identify the compartment.

The conventional TI dose map is an assumption about physiology, not a consequence of field physics. The envelope is a mathematical property of a high-frequency waveform. It is not itself a low-frequency field, and nothing in Maxwell's equations delivers it to a neuron;

biology must construct it. Targeting work since [1] takes the envelope-modulation amplitude $\mathcal{M}[A] = 2\min(\varepsilon_1, \varepsilon_2)$ as the spatial dose, which assumes $R(\mathbf{x}) = \Phi(\mathcal{M}[A])$ for monotonic Φ—that the response depends on the envelope excursion *alone*, and not separately on the carrier baseline, the operating point or the orientation. That assumption is substantive, and a weak smooth nonlinearity cannot satisfy it: the carrier average Eq. (14) is invariant under $A \to -A$, so its expansion contains only even powers and no term proportional to A exists. Square-law transduction is therefore the generic perturbative answer and envelope following is the special case needing justification, since recovering A requires rectification (§2.4) and any *smoothed* threshold reverts to A^2 as the field weakens.

Different field strengths and operating states may recruit different nonlinear mechanisms; the present analysis addresses the weak-response regime. Near threshold a sharp channel or spike nonlinearity may act as a rectifier; well below it, the leading response of any smooth nonlinearity, memoryless or dynamic, is quadratic mixing, filtered and amplified by the network as described here. Which of the two operates at human field strengths is an empirical question that present evidence does not settle. Primate recordings at human-comparable fields find an AM waveform less effective than matched-depth tACS, but do not estimate the difference closely enough to select a transduction law [6], and peripheral-nerve work argues that some TI responses reflect carrier integration rather than envelope extraction [5]. Two 2025 studies sharpen that point at suprathreshold intensities: in peripheral nerves the strength–frequency curves of TI, amplitude-modulated and unmodulated kHz waveforms overlap [4], and in isolated hippocampal neurons pure 2 and 20 kHz fields excite at high intensity while modulation patterns the firing without lowering the threshold [11]. At those intensities the carrier itself stimulates. The weak-field mixing route analyzed here is a candidate for the subthreshold, modulatory regime of the primate recordings and of the recent patient studies of hippocampal TI [6, 39]. What settles it is not whether TI works but which quantity it transduces, and §2.4 gives the measurement.

The amplifier is not specific to TI. A tACS field oscillates in the envelope band directly and needs no demodulation, but it drives the same near-Hopf resonator. Sweeping the coupling toward the alpha Hopf at *pinned* operating point, a direct Δf drive reproduces the same $1/\gamma$ growth and resonance sharpening as the demodulated TI drive, linear in the field rather than square-law (Fig. S3). Coupling-controlled gain is known: the dynamical transfer function of an E–I network is known to become resonant near a Hopf bifurcation, with a peak that *diverges* on the bifurcation line as connectivity is varied [40], and increased self-connectivity is known to enhance weak-field resonant sensitivity in the exact mean field [22]. What the pinned sweep adds is the separation of amplification from *detection*. Holding the local detector curvature $\sigma''(v^*)$ constant while the network transfer changes isolates the network gain, and shows that the same amplifier acts on a drive the population has itself generated. TI then differs from tACS by a generating front end that converts a kHz carrier pair into an effective in-band drive; downstream, both share the criticality- and coupling-controlled amplifier, so, matched for effective in-band drive, read through the same observable and near the same state, they should entrain alike. The cancellation is conditional on those matches.

Amplification is critical slowing, of a mean-field kind. Gain $\propto 1/\gamma$ as the damping vanishes at the Hopf is the diverging susceptibility of a system approaching a bifurcation. In the exact next-generation mean field the reduction from the spiking population is exact in the thermodynamic limit and (r, v) are order parameters [41, 20], so the Hopf is a macroscopic transition; because the reduction assumes all-to-all coupling the critical point is mean-field, with temporal critical slowing but no diverging spatial correlation length. The coupling sweep moves the model toward the bifurcation; the stimulus probes its response.

Prior work already places the operative nonlinearity in the network. Recurrent excitation, with short-term synaptic depression, accounts for tACS entrainment in a large-scale spiking model and yields an Arnold tongue over stimulation amplitude and frequency [42]; the dynami-

cal transfer function of an E–I network, in both rate and spiking form, is resonant near a Hopf bifurcation with a peak that diverges on the bifurcation line as the connectivity is swept [40]; an amplitude-modulated high-frequency drive phase-locks a cortical model to its envelope through the same target engagement as low-frequency tACS [43]; and comparing neuron models directly, integrate-and-fire membranes fail to reproduce observed TI behavior while Hodgkin–Huxley ones succeed [44]. The general case for stimulation as interaction with ongoing dynamics has been made explicitly [45]. What we add is the separation of the generated source from the network response, the comparison of a memoryless and a dynamic detector, and, downstream of both, that the transduced quantity in the weak-response regime is A^2 rather than A, a distinction the cited network studies do not draw.

Estimated response magnitude and comparison with tACS. Two obstacles separate a transcranial field from a response at Δf, and they need different answers. The first is that weak fields barely move a membrane: measured coupling is 0.12–0.21 mV per V/m in hippocampus [46, 47] and up to ∼0.5 mV per V/m for cortical L5/6 pyramidal cells [48], human cortex sees about 0.8 V/m at 2 mA [49, 50], and most applied current is shunted through scalp and skull [51]. The field shifts a soma by a few tenths of a millivolt against the ∼23 mV separating rest from threshold [52]. This is as true of tACS as of TI. The second belongs to TI alone: its field carries no power at Δf, so the response there must be generated by a nonlinearity. Resonance addresses the first, a population near its own bifurcation responding to a weak push far more strongly than an isolated cell, but not the second, because it amplifies TI and tACS alike and cancels when the two are compared under the matching conditions above.

The resonance contributes a factor of order ten. With $\tau = 1/\gamma$ the reverberation time, the gain at resonance relative to the same circuit's response to a steady input is $\mathcal{Q} = \pi f_0 \tau$. Cortical population activity relaxes over a few hundred milliseconds [53], giving $\mathcal{Q}$ of order 10 at $f_0 = 10$ Hz—an order of magnitude, not a value, since that timescale is wideband rather than alpha-specific. Gain also costs bandwidth: the resonance is $1/(\pi\tau)$ wide in power (full width at half maximum), about 1 Hz, so a Δf fixed at a group-mean alpha may sit outside it; whether an individual's alpha drifts that far within a session is not settled by this estimate. At equal field strength TI is far weaker than tACS for the memoryless detector at matched peak polarization. With per-carrier polarizations $\varepsilon_{1,2}$ the demodulated drive is $\frac{1}{2}\sigma''\varepsilon_1\varepsilon_2$ against $2\sigma'\varepsilon$ for tACS at the same peak polarization, a ratio $\eta_{\mathrm{det}} = \sigma''\varepsilon/4\sigma'$ proportional to ε: a square-law detector gets *less* efficient as the signal weakens. Formally η_{det} reaches unity near $\varepsilon \approx 7$ mV, but that is the order of the distance to threshold and lies outside the expansion's validity, so parity is not reachable by this route rather than merely expensive. At 1 V/m with a 1 kHz carrier—now comparing at matched *applied* field, so the carrier's membrane attenuation enters twice while the in-band tACS drive is unfiltered—TI comes out ∼300× weaker than tACS if the carrier reaches a fast nodal compartment, and $\sim 10^6\times$ weaker if it reaches only the soma.

The one in-regime measurement sits well above this estimate. Vieira et al. worked at modulation depths of 0.5–0.62 V/m, chosen deliberately to match human studies rather than the 60–383 V/m of the rodent literature, and report TI at 0.34 times the strength of conventional tACS (95% CI 0.04–0.65), or ∼12% per unit current under their stated assumption that effects are roughly proportional to applied current [6]. That normalization is not neutral here, since a square-law drive scales as I^2 while tACS scales as I; the per-current figure is therefore transduction-class-dependent and we do not use it. Their AM-versus-tACS comparison is the most direct available. Having measured that transcranial current is shunted increasingly with frequency, they compensate for it and deliver conventional tACS and an AM waveform through the same electrodes, with the modulation depth near the recorded cells calibrated to 1.14–1.17 times the tACS amplitude. Against a baseline PLV of 0.043, tACS raises the median to 0.20 (95% CI 0.043–0.64) and AM-tACS to 0.10 (0.023–0.38), a paired difference significant at $p = 0.01$.

Taking their fields with a favorable cortical coupling (∼0.5 mV per V/m) through a fast element at

2 kHz gives $\varepsilon \approx 0.1$ mV and hence $\eta \approx 0.8\%$, far below the ordering they report; a somatic route is worse by two further orders. Three limits keep this from being a quantitative test. The comparison rests on twenty neurons, of which ten responded to at least one modality and seven to both, so the sample is selected for responsiveness and the reported intervals are correspondingly broad. PLV is a downstream statistic that depends on baseline locking, network susceptibility and state as well as on the generated drive, and in their full dataset baseline PLV predicts the sign as well as the size of the change. And the quantity matched was the extracellular modulation, not the post-coupling polarization $\varepsilon = \lambda_f E$ of §2.2 at whichever compartment supplies the nonlinearity, so the ratio reports the whole end-to-end path from applied field through frequency-dependent access, transduction, network dynamics and spike-phase statistics. It does not isolate the transducer. What the experiment establishes is that a high-frequency AM waveform drives phase locking less effectively than a matched-depth low-frequency one, and that the prominent population effect was often *reduced* locking to ongoing activity rather than increased beat locking, a phenotype richer than the increased envelope-locked modulation our models show; the magnitude of that difference is not estimated precisely enough to discriminate among candidate transduction laws.

For a memoryless transfer the ratio is bounded whatever the operating point: by $\rho\varepsilon/4$ at fixed slope, and by 0.348 over the whole logistic family, a ceiling approached only where the transfer has narrowed to a microvolt width (App. D). Brain state moves the efficiency but cannot lift it off that ceiling.

A sigmoid steep enough to approach the ceiling is a step detector, which is neither the square law nor the linear envelope follower of §2.4: its harmonic ratio at the optimum is 0.15, not 1/5 (App. D). Which transduction operates in vivo is not settled by the evidence available. The ceiling is a supremum over a family of memoryless transfers at matched peak polarization, not a prediction for any one preparation, and it does not govern the dynamic detector of §2.3; its numerical proximity to a reported ratio is neither support nor exclusion. Vieira et al. read their own data as favoring rectification, invoking the radio analogy explicitly and proposing asymmetry between inward Na^+ and outward K^+ channels as the operative nonlinearity. They also note that circuit-level frequency dependence could contribute to demodulation, which is the mechanism developed here.

A retrospective test on existing primate data would need the link between generated drive and the phase-locking value (PLV) fixed or calibrated: with an unspecified monotone link the amplitude exponent is not identifiable, since the link absorbs it. Prospectively, matching post-coupling polarization rather than extracellular field alone, and reading the amplitude slope together with the transfer-corrected harmonic ratio of §2.4, separates the effective response laws without an absolute calibration.

The amplification half is separate, for a definite reason. The J-curve reports A_Ω/A_{open}, the closed-loop response normalized by the open-loop detector output at the same operating point. The detector divides out by construction: whatever manufactures the beat component—a smooth population curvature, a nodal rectifier, a spike threshold—the recurrent network's response to that in-band drive is the same, and it is that response which follows $1/\gamma$ and sharpens toward the bifurcation. The same holds in the exact mean field, where the rectifier is the derived v_E^2 term and no operating-point pinning is needed at all. Detection and amplification are therefore separable in experiment as well as in principle, and today's evidence bears only on the first. The population contribution established here is the tuning and amplification; detection may occur microscopically or through the effective population transfer.

Whether square-law transduction operates in the human brain at safe field strengths is open. What the parity argument establishes is narrower: any smooth weak nonlinearity with a nonzero second-order response transduces A^2, so a square-law route exists and is the generic leading perturbative form, not a quantitative estimate of the in-vivo response. The evidence available is consistent with a more rectifying transduction in vivo, in which case the detector is cellular; it does not establish

that. The amplification half is unaffected either way: whatever generates the beat component, a network near its bifurcation amplifies it as $1/\gamma$. The amplitude sweep of §2.4 distinguishes these effective response laws under the stated controls: a slope of 1 with a harmonic ladder near 1/5 indicates envelope following, a slope of 2 with a vanishing ladder the square law, and a step-like threshold gives intermediate ratios (App. D).

Limitations. The field enters the population transfer directly; we treat the field-to-nonlinearity coupling λ_f phenomenologically rather than modeling channel kinetics, and ε should be read as post-coupling polarization. The population sigmoid is a coarse-grained stand-in for distributed single-cell nonlinearities. JR alpha near this Hopf is high quality-factor, so the stable-side peaks are narrow and would be broadened by noise and population heterogeneity; the columns here are single, deterministic and noiseless, and coupled noisy columns with an explicit access filter $\lambda_f(f_c)$ remain the natural next step. Cascading ordinary neural-mass transfers does not solve carrier access, since the intervening synaptic filter removes the kHz component before a second sigmoid can rectify it. A population of microscopic rectifiers could supply the effective detector, but heterogeneity in their thresholds would broaden the aggregate transfer, and whether it stays sharp enough at human fields is a separate biophysical question. The exact mean field resolves the static-versus-dynamic question for a QIF population with an instantaneous quadratic; real spike initiation has gate kinetics of 0.1–0.2 ms, so at 1 kHz the gate is not a small parameter, and the fast-carrier coefficient of §2.3 is a model statement, not a measured cellular one; single-compartment modeling finds that channel gating kinetics, not the current–voltage nonlinearity, carry the TI response, and that integrate-and-fire cells fail to reproduce it [44]. That negative result concerns an isolated cell, whose slow stage must be intracellular. In a population the synaptic and recurrent dynamics supply the slow stage, and the cell need only rectify the carrier: a network of quadratic integrate-and-fire cells responds to TI (§4.8) with the spike-initiation quadratic as the rectifier, and a population of hard-threshold cells with spread thresholds would respond through the fraction crossing threshold at the carrier peaks, linearly in the envelope above the spread and quadratically below it. The human evidence is also not uniformly positive: a controlled study found peripheral but no central effects [54], and a systematic review reports that occipito-parietal TI did not modulate alpha power [55]. The narrowband prediction turns the second into a design question—whether Δf tracked individual alpha closely enough to stay inside a $\sim$1 Hz response band—which is a prediction to be tested, not an explanation to be assumed.

6 Conclusion

The models show how neural populations can act as tuned receivers of temporal-interference inputs. Nonlinear mixing generates a component at the difference frequency, and recurrent dynamics selectively amplify it near a network resonance. Generation and amplification are distinct contributions: a large network response does not, by itself, imply efficient detection.

The detector's dynamics matter. A memoryless firing-rate sigmoid and a dynamic quadratic-integrate-and-fire mean field both support quadratic mixing in the weak-response regime, but their mixing coefficients and operating-point dependence differ. A steady-state firing-rate curve therefore does not, on its own, determine the response to a fast carrier; the crossover from curvature to slope sits at the population's own frequency, below the carriers TI uses. The common mechanism is nonlinear mixing followed by network tuning, not a universal sigmoid-curvature law.

This distinction changes how the stimulation should be interpreted. In the local square-law description, beat generation depends on the product of the two effective carrier inputs rather than only on the envelope excursion, and network susceptibility then determines how strongly that generated component appears in the response. When TI and a direct low-frequency drive excite the same effective input and observable around the same state, they share the downstream amplifier, so resonance alone does not give TI an efficiency advantage. Which compartment passes a kHz carrier to the mixing nonlinearity is a hypothesis in an access-filter model, and the absolute re-

sponse magnitude at human field strengths remains to be established. The analysis identifies what should be measured separately: carrier access, nonlinear beat generation, and circuit tuning.

Code and data availability

All simulation and figure-generation code, the committed datasets, and a one-command reproduce script (`run_all.py`) are openly available at `https://github.com/giulioruffini/TI-demodulation-paper`, archived at Zenodo (doi:10.5281/zenodo.21009618). The repository `README` documents the per-figure script mapping (a figure→script table) and the dependency order, so every figure in this note can be regenerated from the committed inputs. The code is released under CC-BY-4.0.

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Competing interests

GR is Chief Scientific and Technology Officer of Neuroelectrics, which develops and commercializes transcranial stimulation devices. BM, AJ and FC are affiliated with Neuroelectrics. The remaining authors declare no competing interests.

Author contributions

Using the CRediT taxonomy: **G. Ruffini** — conceptualization, methodology, formal analysis, software, investigation, visualization, writing (original draft), supervision, funding acquisition. **B. Mercadal** — validation, visualization, writing (review and editing). **A. Just** — software, visualization, writing (review and editing). **R. de Palma Aristides** — software, visualization, writing (review and editing). **F. Castaldo** — conceptualization, writing (review and editing). **S. Canals** — supervision, writing (review and editing). **C. Mirasso** — supervision, writing (review and editing).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic Claude and OpenAI ChatGPT to streamline the narrative and to assist with code and figure generation from data the authors produced. The authors reviewed and edited all resulting content and take full responsibility for the content of the published article.

References

[1] Nir Grossman, David Bono, Nina Dedic, Suhasa B. Kodandaramaiah, Andrii Rudenko, Ho-Jun Suk, Antonino M. Cassara, Esra Neufeld, Niels Kuster, Li-Huei Tsai, Alvaro Pascual-Leone, and Edward S. Boyden. Noninvasive deep brain stimulation via temporally interfering electric fields. *Cell*, 169(6):1029–1041.e16, 2017.

[2] Ehsan Mirzakhalili, Beatrice Barra, Marco Capogrosso, and Scott F. Lempka. Biophysics of temporal interference stimulation. *Cell Systems*, 11(6):557–572.e5, 2020.

[3] Paul J. Nahin. *The Science of Radio: with MATLAB and Electronics Workbench Demonstrations*. Springer, 2nd edition, 2001.

[4] Aleksandar Opančar, Petra Ondráčková, David Samuel Rose, Jan Trajlinek, Vedran Đerek, and Eric Daniel Głowacki. The same biophysical mechanism is involved in both temporal interference and direct kHz stimulation of peripheral nerves. *Nature Communications*, 16:9006, 2025.

[5] Ryan B. Budde, Michael T. Williams, and Pedro P. Irazoqui. Temporal interference current stimulation in peripheral nerves is not driven by envelope extraction. *Journal of Neural Engineering*, 20(2):026041, 2023.

[6] Pedro G. Vieira, Matthew R. Krause, and Christopher C. Pack. Temporal interference stimulation disrupts spike timing in the primate brain. *Nature Communications*, 15:4558, 2024.

[7] Sara Caldas-Martinez, Chaitanya Goswami, Mats Forssell, Jiaming Cao, Alison L. Barth, and Pulkit Grover. Cell-specific effects of temporal interference stimulation on cortical function. *Communications Biology*, 7:1076, 2024.

[8] Nafiseh Karimi, Rassoul Amirfattahi, and Abolghasem Zeidaabadi Nezhad. Neuromodulation effect of temporal interference stimulation based on network computational model. *Frontiers in Human Neuroscience*, 18, 2024.

[9] Hossein Soroushi, Sara Abbasi, Yun Du, Ning Ning, and Yuming Lei. Temporal interference stimulation: mechanisms, optimization, validation, and clinical prospects—a comprehensive review. *WIREs Computational Statistics*, 17(3):e70031, 2025.

[10] Wei Wang, M. Florian, C. Ling, K. Wei, J. Jian, H. Sun, L. Yan, S. Hu, and B. Bi. Temporal interference stimulation: a new paradigm for non-invasive deep brain stimulation—mechanistic insights, clinical advances, and future directions. *Translational Psychiatry*, 2026.

[11] Iurii Semenov, Vitalii Kim, Giedre Silkuniene, and Andrei G. Pakhomov. Excitation and polarization of isolated neurons by high-frequency sine waves for temporal interference stimulation. *Cell Reports Physical Science*, 2025.

[12] Giulio Ruffini, Edmundo Lopez-Sola, Raul Palma, Roser Sanchez-Todo, Jakub Vohryzek, Francesca Castaldo, and Karl Friston. Cross-frequency coupling as a neural substrate for prediction error evaluation: a laminar neural mass modeling approach. *bioRxiv*, 2025. Preprint 2025.03.19.644090.

[13] Giulio Ruffini. Neural encoding through hierarchical amplitude modulation. *bioRxiv*, 2025. Preprint 2025.11.03.686310.

[14] Francesca Castaldo, Raul de Palma Aristides, Pau Clusella, Jordi Garcia-Ojalvo, and Giulio Ruffini. Rosetta Stone of neural mass models. *Physics Reports*, 1189:1–49, 2026. arXiv:2512.10982.

[15] Ben H. Jansen and Vincent G. Rit. Electroencephalogram and visual evoked potential generation in a mathematical model of coupled cortical columns. *Biological Cybernetics*, 73(4):357–366, 1995.

[16] Hugh R. Wilson and Jack D. Cowan. Excitatory and inhibitory interactions in localized populations of model neurons. *Biophysical Journal*, 12(1):1–24, 1972.

[17] Bruce Hutcheon and Yosef Yarom. Resonance, oscillation and the intrinsic frequency preferences of neurons. *Trends in Neurosciences*, 23(5):216–222, 2000.

[18] Nicolas Brunel and Vincent Hakim. Fast global oscillations in networks of integrate-and-fire neurons with low firing rates. *Neural Computation*, 11(7):1621–1671, 1999.

[19] Nicolas Brunel and Xiao-Jing Wang. What determines the frequency of fast network oscillations with irregular neural discharges? I. synaptic dynamics and excitation–inhibition balance. *Journal of Neurophysiology*, 90(1):415–430, 2003.

[20] Ernest Montbrió, Diego Pazó, and Alex Roxin. Macroscopic description for networks of spiking neurons. *Physical Review X*, 5(2):021028, 2015.

[21] Giulio Ruffini. Analysis and extension of exact mean-field theory with dynamic synaptic currents. *bioRxiv*, 2021. Preprint 2021.09.01.458563 (NMM2).

[22] Pau Clusella, Elif Köksal-Ersöz, Jordi Garcia-Ojalvo, and Giulio Ruffini. Comparison between an exact and a heuristic neural mass model with second-order synapses. *Biological Cybernetics*, 117(1-2):5–19, 2023.

[23] Roser Sanchez-Todo, André M. Bastos, Edmundo Lopez-Sola, Borja Mercadal, Emiliano Santarnecchi, Earl K. Miller, Gustavo Deco, and Giulio Ruffini. A physical neural mass model framework for the analysis of oscillatory generators from laminar electrophysiological recordings. *NeuroImage*, 270:119938, 2023.

[24] Adrià Galan-Gadea, Ricardo Salvador, Fabrice Bartolomei, Fabrice Wendling, and Giulio Ruffini. Spherical harmonics representation of the steady-state membrane potential shift induced by tDCS in realistic neuron models. *Journal of Neural Engineering*, 20(2):026004, 2023.

[25] Yair Deitcher, Guy Eyal, Lida Kanari, Matthijs B. Verhoog, Guy Antoine Atenekeng Kahou, Huibert D. Mansvelder, Christiaan P. J. de Kock, and Idan Segev. Comprehensive morpho-electrotonic analysis shows 2 distinct classes of L2 and L3 pyramidal neurons in human temporal cortex. *Cerebral Cortex*, 27(11):5398–5414, 2017.

[26] Guy Eyal, Matthijs B. Verhoog, Guilherme Testa-Silva, Yair Deitcher, Johannes C. Lodder, Ruth Benavides-Piccione, Juan Morales, Javier DeFelipe, Christiaan P. J. de Kock, Huibert D. Mansvelder, and Idan Segev. Unique membrane properties and enhanced signal processing in human neocortical neurons. *eLife*, 5:e16553, 2016.

[27] Davide Reato, Asif Rahman, Marom Bikson, and Lucas C. Parra. Low-intensity electrical stimulation affects network dynamics by modulating population rate and spike timing. *Journal of Neuroscience*, 30(45):15067–15079, 2010.

[28] Anli Liu, Mihály Vöröslakos, Greg Kronberg, Simon Henin, Matthew R. Krause, Yu Huang, Alexander Opitz, Ashesh Mehta, Christopher C. Pack, Bart Krekelberg, Antal Berényi, and Lucas C. Parra. Immediate neurophysiological effects of transcranial electrical stimulation. *Nature Communications*, 9:5092, 2018.

[29] François Grimbert and Olivier Faugeras. Bifurcation analysis of Jansen's neural mass model. *Neural Computation*, 18(12):3052–3068, 2006.

[30] Martin Vinck, Marijn van Wingerden, Thilo Womelsdorf, Pascal Fries, and Cyriel M. A. Pennartz. The pairwise phase consistency: a bias-free measure of rhythmic neuronal synchronization. *NeuroImage*, 51(1):112–122, 2010.

[31] Arkady Pikovsky, Michael Rosenblum, and Jürgen Kurths. *Synchronization: A Universal Concept in Nonlinear Sciences*. Cambridge University Press, 2001.

[32] Ryan T. Canolty and Robert T. Knight. The functional role of cross-frequency coupling. *Trends in Cognitive Sciences*, 14(11):506–515, 2010.

[33] Adriano B. L. Tort, Robert Komorowski, Howard Eichenbaum, and Nancy Kopell. Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *Journal of Neurophysiology*, 104(2):1195–1210, 2010.

[34] Zeinab Esmaeilpour, Mark Jackson, Greg Kronberg, Tianhe Zhang, Rosana Esteller, Brad Hershey, and Marom Bikson. Limited sensitivity of hippocampal synaptic function or network

oscillations to unmodulated kilohertz electric fields. *eNeuro*, 7(6):ENEURO.0368–20.2020, 2020.

[35] Boshuo Wang, Aman S. Aberra, Warren M. Grill, and Angel V. Peterchev. Responses of model cortical neurons to temporal interference stimulation and related transcranial alternating current stimulation modalities. *Journal of Neural Engineering*, 19(6):066047, 2022.

[36] Asif Rahman, Davide Reato, Mattia Arlotti, Fernando Gasca, Abhishek Datta, Lucas C. Parra, and Marom Bikson. Cellular effects of acute direct current stimulation: somatic and synaptic terminal effects. *The Journal of Physiology*, 591(10):2563–2578, 2013.

[37] Aman S. Aberra, Ruochen Wang, Warren M. Grill, and Angel V. Peterchev. Multi-scale model of axonal and dendritic polarization by transcranial direct current stimulation in realistic head geometry. *Brain Stimulation*, 16(6):1776–1791, 2023.

[38] Darpan Chakraborty, Dennis Q. Truong, Marom Bikson, and Hanoch Kaphzan. Neuromodulation of axon terminals. *Cerebral Cortex*, 28(8):2786–2794, 2018.

[39] Florian Missey, Emma Acerbo, Adam S. Dickey, et al. Non-invasive temporal interference stimulation of the hippocampus suppresses epileptic biomarkers in patients with epilepsy. *Brain Stimulation*, 19(1):102981, 2026.

[40] Erwan Ledoux and Nicolas Brunel. Dynamics of networks of excitatory and inhibitory neurons in response to time-dependent inputs. *Frontiers in Computational Neuroscience*, 5, 2011.

[41] Edward Ott and Thomas M. Antonsen. Low dimensional behavior of large systems of globally coupled oscillators. *Chaos*, 18(3):037113, 2008.

[42] M. M. Ali, K. K. Sellers, and F. Frohlich. Transcranial alternating current stimulation modulates large-scale cortical network activity by network resonance. *Journal of Neuroscience*, 33(27):11262–11275, 2013.

[43] Ehsan Negahbani, Florian H. Kasten, Christoph S. Herrmann, and Flavio Fröhlich. Targeting alpha-band oscillations in a cortical model with amplitude-modulated high-frequency transcranial electric stimulation. *NeuroImage*, 173:3–12, 2018.

[44] Thomas Plovie, Ruben Schoeters, Thomas Tarnaud, Wout Joseph, and Emmeric Tanghe. Nonlinearities and timescales in neural models of temporal interference stimulation. *Bioelectromagnetics*, 46(1):e22522, 2025.

[45] Matthew R. Krause, Pedro G. Vieira, and Christopher C. Pack. Transcranial electrical stimulation: How can a simple conductor orchestrate complex brain activity? *PLOS Biology*, 21(1):e3001973, January 2023.

[46] Marom Bikson, Masashi Inoue, Hiroki Akiyama, Jackie K. Deans, John E. Fox, Hiroyoshi Miyakawa, and John G. R. Jefferys. Effects of uniform extracellular dc electric fields on excitability in rat hippocampal slices in vitro. *The Journal of Physiology*, 557(1):175–190, May 2004.

[47] Jacqueline K. Deans, Andrew D. Powell, and John G. R. Jefferys. Sensitivity of coherent oscillations in rat hippocampus to ac electric fields. *The Journal of Physiology*, 583(2):555–565, August 2007.

[48] Thomas Radman, Raddy L. Ramos, Joshua C. Brumberg, and Marom Bikson. Role of cortical cell type and morphology in subthreshold and suprathreshold uniform electric field stimulation in vitro. *Brain Stimulation*, 2(4):215–228.e3, October 2009.

[49] Yu Huang, Anli A Liu, Belen Lafon, Daniel Friedman, Michael Dayan, Xiuyuan Wang, Marom Bikson, Werner K Doyle, Orrin Devinsky, and Lucas C Parra. Measurements and models of

electric fields in the in vivo human brain during transcranial electric stimulation. *eLife*, 6, February 2017.

[50] Yu Huang, Anli A Liu, Belen Lafon, Daniel Friedman, Michael Dayan, Xiuyuan Wang, Marom Bikson, Werner K Doyle, Orrin Devinsky, and Lucas C Parra. Correction: measurements and models of electric fields in the in vivo human brain during transcranial electric stimulation. *eLife*, 7, February 2018.

[51] Mihály Vöröslakos, Yuichi Takeuchi, Kitti Brinyiczki, Tamás Zombori, Azahara Oliva, Antonio Fernández-Ruiz, Gábor Kozák, Zsigmond Tamás Kincses, Béla Iványi, György Buzsáki, and Antal Berényi. Direct effects of transcranial electric stimulation on brain circuits in rats and humans. *Nature Communications*, 9(1), February 2018.

[52] Kohei Koga, Xiangyao Li, Tao Chen, Hendrik W Steenland, Giannina Descalzi, and Min Zhuo. In vivo whole-cell patch-clamp recording of sensory synaptic responses of cingulate pyramidal neurons to noxious mechanical stimuli in adult mice. *Molecular Pain*, 6, January 2010.

[53] Jens Wilting and Viola Priesemann. Inferring collective dynamical states from widely unobserved systems. *Nature Communications*, 9(1), 2018.

[54] Krisztián Iszak, Simon Mathies Gronemann, Stefanie Meyer, Alexander Hunold, Jana Zschüntzsch, Mathias Bähr, Walter Paulus, and Andrea Antal. Why temporal inference stimulation may fail in the human brain: A pilot research study. *Biomedicines*, 11(7):1813, 2023.

[55] Paria Mansourinezhad, Rob M C Mestrom, Debby C W Klooster, Mathieu Sprengers, Paul A J M Boon, and Margarethus M Paulides. Systematic review of experimental studies in humans on transcranial temporal interference stimulation. *Journal of Neural Engineering*, 22(5):051001, 2025.